TCS NQT 2027 · REASONING ABILITY · COMPLETE MASTERY GUIDE

Reasoning Ability Complete Mastery

Beginner to advanced: every reasoning topic explained with diagrams, formulas, solved examples, tricks and practice sets. All numbers checked by computer.

20	76	155	40
chapters, basic to advanced	solved examples	practice and test questions	diagrams drawn for you

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About this guide

The reasoning section of TCS NQT and the Advanced test checks logic and speed. This guide teaches every question type from zero: the idea in simple words, the formulas, diagrams, solved examples, shortcuts, common mistakes and graded practice (Basic, Medium, Advanced).

How each chapter is built

- **Idea:** what the topic is in plain words.
- **Diagram:** a picture of the method.
- **Formulas and rules:** the yellow box to revise.
- **Solved examples** with full steps.
- **Shortcuts and mistakes** to avoid.
- **Practice set** with answers at the end.

How to study

- Read one chapter and redo the solved examples without looking.
- Solve the practice set, then check the answer key.
- Keep an error log of wrong answers.
- Revise the formula boxes every day.
- Finish with the mini test and the formula sheet in the last chapter.

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Honest note

- All numerical answers, seating and puzzle solutions, syllogism conclusions, counts and dice/net facts in this guide were verified by running code.
- Verbal reasoning (statement-based) follows standard exam conventions; where textbooks differ the guide says so.

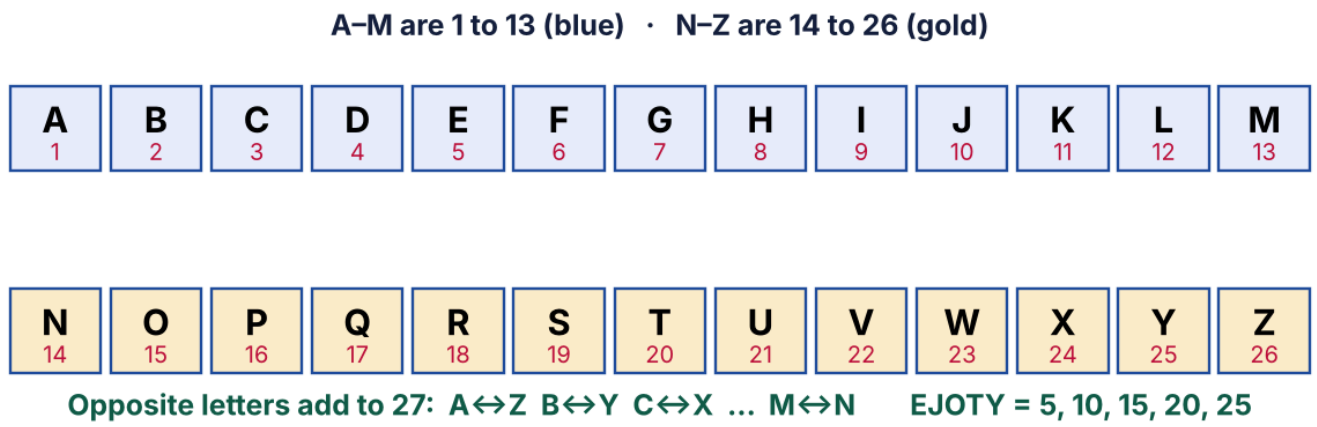
01 Number and Alphabet Series

REASONING ABILITY · PART A · MUST-DO

A series follows a hidden rule. You see the first few terms, discover the rule, and apply it once more. It is the fastest scoring topic: the question is short and almost every rule is one of about fifteen patterns that you can learn once and recognise forever.

Step 1: learn the letter positions

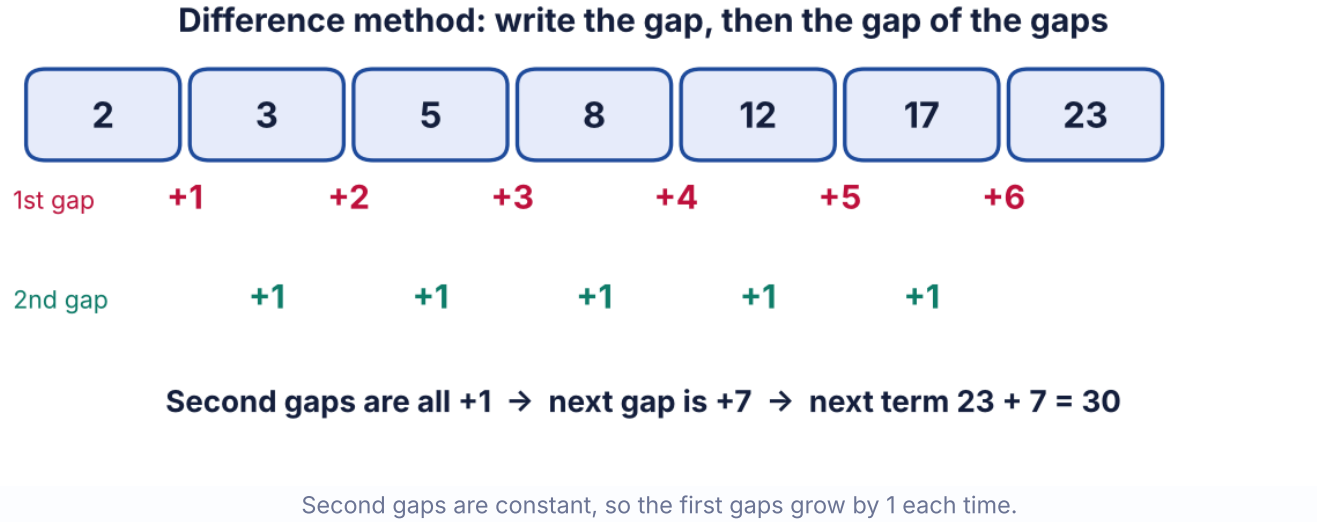
Every alphabet question becomes a number question once you convert letters to positions. Learn the picture below; also remember the anchor letters **E=5, J=10, O=15, T=20, Y=25** so you can count forward or backward without writing the full alphabet.



Alphabet with positions. The opposite of a letter is 27 minus its position.

Step 2: find the rule — the difference method

Write the gap between neighbouring terms. If the gaps are equal the series is arithmetic. If the gaps themselves change, write the gaps of the gaps (second row). Most TCS series are solved in two rows.

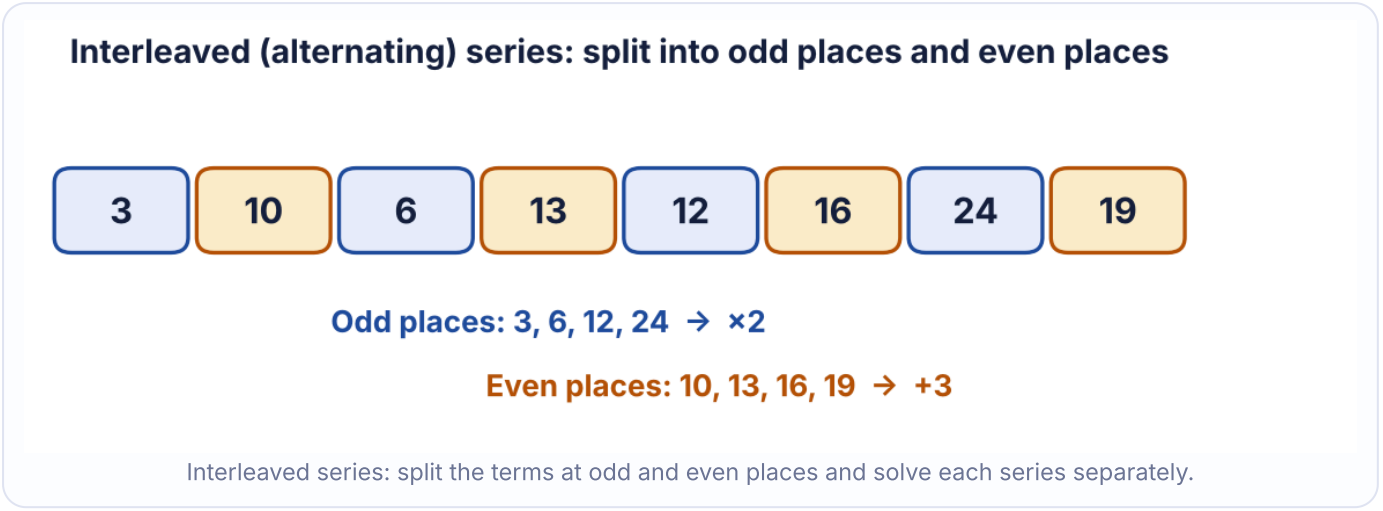


Second gaps are constant, so the first gaps grow by 1 each time.

The pattern table (learn this once)

Type	Form	Example → next	How to spot

Arithmetic	a, a+d, a+2d ...	4, 9, 14, 19 → 24	add the same number
Geometric	a, ar, ar^2 ...	3, 6, 12, 24 → 48	multiply by the same number
Squares / cubes	n^2, n^3	1, 4, 9, 16 → 25 ; 1, 8, 27, 64 → 125	compare with perfect squares/cubes
Square ± k	$n^2 \pm k$	2, 5, 10, 17 → 26 (n^2+1)	nearest square gives the clue
$n^2 + n$	n(n+1)	2, 6, 12, 20 → 30	product of consecutive numbers
$n^3 \pm n$ or $n^3 \pm 1$		0, 6, 24, 60 → 120 (n^3−n)	near cubes
Growing gaps	gap +1 each time	1, 2, 4, 7, 11 → 16	second row of gaps constant
×k + c	a(n+1)=k·a(n)+c	3, 7, 15, 31 → 63 (×2+1)	look at ratio ≈ k
Prime numbers	2,3,5,7,11,13...	2, 3, 5, 7, 11 → 13	irregular small gaps
Fibonacci type	a(n)=a(n−1)+a(n−2)	2, 3, 5, 8, 13 → 21	each term = sum of two before it
Alternating	+a, −b, +a, −b	5, 9, 7, 11, 9 → 13	two moves repeat
Interleaved	two series woven	3, 10, 6, 13, 12, 16 → 24	split odd/even places
Multiply by 1,2,3...	×1, ×2, ×3	2, 2, 4, 12, 48 → 240	multiplier also grows
Add squares / cubes	gaps are n^2	1, 2, 6, 15, 31 → 56	gaps 1, 4, 9, 16, 25
Digit based	sum/product of digits	12, 24, 36 ...	look inside the number



Interleaved series: split the terms at odd and even places and solve each series separately.

Formulas to keep

- Arithmetic: $a_n = a + (n-1)d$ · Geometric: $a_n = a \cdot r^{n-1}$
- Sum of first n numbers: **n(n+1)/2** (triangular numbers 1, 3, 6, 10, 15 ...)
- Sum of first n odd numbers = n^2; sum of first n squares = **n(n+1)(2n+1)/6**
- Position from the other end: **27 − position**; opposite letter pairs sum to 27
- Letters in a cluster series: write each letter as a number and treat each place as its own series

Alphabet and alphanumeric series

Skip patterns. A C E G (+2), Z X V T (−2), B E H K (+3). **Growing skips.** A B D G K (gaps 1, 2, 3, 4). **Reverse + forward.** Z A Y B X (two series).

Clusters. ABC, DEF, GHI → write numbers (1 2 3, 4 5 6, 7 8 9) and see that each cluster adds 3 to every letter.

Alphanumeric. B2D, E5H, H11L: the first letters move +3, the numbers follow ×2+1 (2, 5, 11, 23), and the last letters move +4. Always split such a term into its parts.

Odd one out. In a list of clusters, find the one whose internal gaps differ from the rest.

Find the next term: 2, 3, 5, 8, 12, 17, ?

23

Gaps are 1, 2, 3, 4, 5, so the next gap is 6: 17 + 6 = 23.

Find the next term: 5, 6, 14, 45, 184, ?

925

5×1+1 = 6, 6×2+2 = 14, 14×3+3 = 45, 45×4+4 = 184, so 184×5+5 = 925.

Find the next letter: C, D, F, I, M, ?

R

Positions 3, 4, 6, 9, 13 have gaps 1, 2, 3, 4; the next gap is 5: 13 + 5 = 18 = R.

Find the next term: 1, 2, 6, 15, 31, ?

56

Gaps are 1, 4, 9, 16 (perfect squares); the next gap is 25: 31 + 25 = 56.

Shortcuts and tricks

- Always compute the gaps first; if they are not simple, try ratios; then try squares and cubes; then split into two series.
- If a number appears like 1, 4, 9, 16 hidden in the gaps, it is a squares pattern.
- For long alphabet jumps, count with E, J, O, T, Y as anchors.
- In a series with one wrong term, find the pattern from the other terms first.

Common mistakes

- Starting from the biggest numbers — look at the first three terms before the rest.
- Forgetting that the alphabet is not circular unless the question says so.
- Using only one gap row when two are needed.

Practice set: Number and alphabet series

1.	3, 8, 18, 38, 78, ?	Basic
2.	4, 9, 25, 49, 121, ?	Medium
3.	Z, X, U, Q, ?	Medium
4.	0, 6, 24, 60, 120, ?	Advanced
5.	B2D, E5H, H11L, K23P, ?	Advanced
6.	2, 5, 12, 27, 58, ?	Medium
7.	AZ, BY, CX, DW, ?	Basic
8.	1, 3, 7, 15, 31, ?	Basic

REASONING ABILITY · PART A · MUST-DO

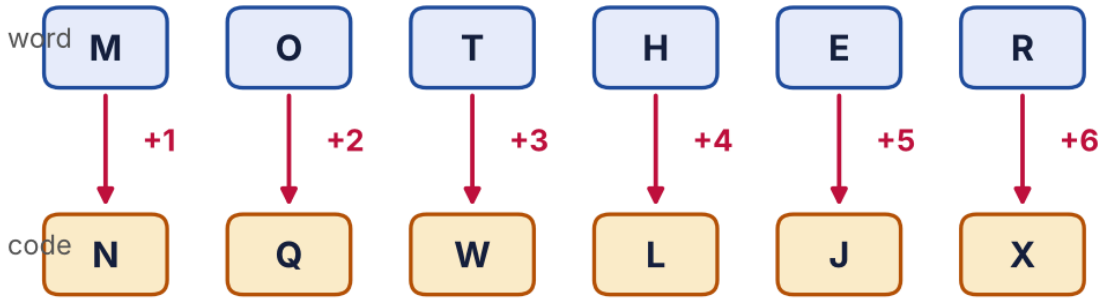
02 Coding and Decoding

A code is a rule that changes words, letters or numbers into other symbols. In the exam you are shown one or two examples and asked to apply the same rule to a new word. The trick is to **find the rule on the first example by comparing letter by letter**, test it on the second example, and then use it.

The seven families of coding

#	Family	Rule	Example
1	Fixed shift	every letter moves +k or −k	CAT → ECV (+2)
2	Progressive shift	shifts form a pattern: +1, +2, +3 ...	MOTHER → NQWLJX
3	Alternating shift	+a, −b, +a, −b ...	DRIVE → FOLRH (+2, −3, +3, −4, +3)
4	Opposite letter	A↔Z, B↔Y ... (sum = 27)	CAT → XZG
5	Reverse + shift	write backwards, then shift	TABLE → GNDCV (reverse, +2)
6	Position / value	letter → its number, sum or product	CAT = 3+1+20 = 24
7	Word / number substitution	each word or digit has a code	find the common word

Letter shift coding: each letter moves a fixed number of steps



MOTHER → NQWLJX: shifts +1, +2, +3, +4, +5, +6 (progressive shift)

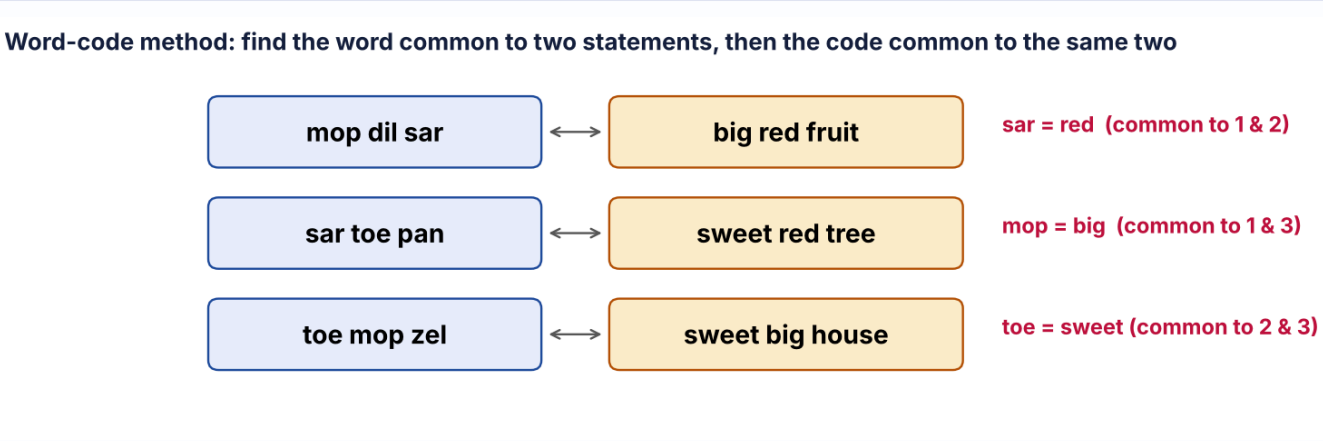
Progressive shift: compare each pair of letters to read the shift (+1, +2, +3 ...).

Formulas and rules

- Shift: new position = old position + k (wrap round after Z: Z + 1 = A)
- Opposite letter of a letter at position p is **27 − p** (A↔Z, M↔N)
- Word value (sum code) = Σ positions; product code = Π positions
- Reverse alphabet position = 27 − normal position
- If the word has more letters than the example, repeat the shift pattern from the start
- Mirror pairs: ABC → ZYX; EJOTY ↔ VQLGB

Word and sentence codes

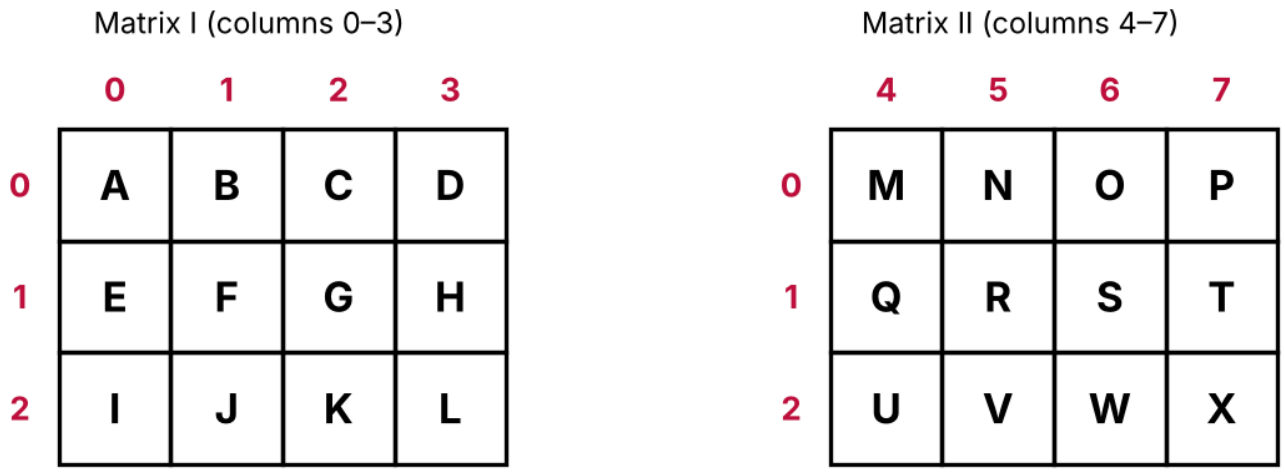
Write each statement as a row. A word that appears in two statements has a code that also appears in both. Cross the pair out and read the next word. This also solves data sufficiency questions on codes.



Common-word method: sar is in statements 1 and 2, and so is "red". So sar = red.

Matrix (table) coding

A letter is written by its row and column numbers. The question usually says whether the row number comes first. In the picture, A = 00, N = 05 (row 0, column 5), T = 17, X = 27. So the word MATE becomes 04, 00, 17, 10.



Two matrices: Matrix I columns are 0–3, Matrix II columns are 4–7.

If LOGIC is coded as NQIKE, how is BRAIN coded?

DTCKP

Each letter moves +2: L→N, O→Q, G→I, I→K, C→E. So B→D, R→T, A→C, I→K, N→P.

If B = 4 and F = 12, find the value of PUT (value of a letter = 2 × position, word = product).

53760

P = 32, U = 42, T = 40. Product = 32 × 42 × 40 = 53760.

If DOG = 104 and EAR = 120, find FISH (word = (sum of positions) × (position of first letter)).

252

DOG: (4+15+7) × 4 = 26 × 4 = 104. FISH: (6+9+19+8) × 6 = 42 × 6 = 252.

In a code, TABLE is written as GZYOV. How is CHAIR written?

XSZRI

Every letter is replaced by its opposite (27 − position): C→X, H→S, A→Z, I→R, R→I.

Shortcuts and tricks

- Compare the first letters of the example and the code; if they differ by 2, test +2 on the next pair.
- If every shift is a different number, look for +1, +2, +3 or an alternating pair.
- For a reversed code, write the code from the right and check opposite letters or a shift.
- In number-letter codes, find which letters share a digit (same digit = same letter).

Common mistakes

- Forgetting to wrap after Z (Y + 3 = B).
- Applying the shift of the example to the wrong letter when the new word is longer.
- Reading the row and the column in the wrong order in a matrix code.

Practice set: Coding practice

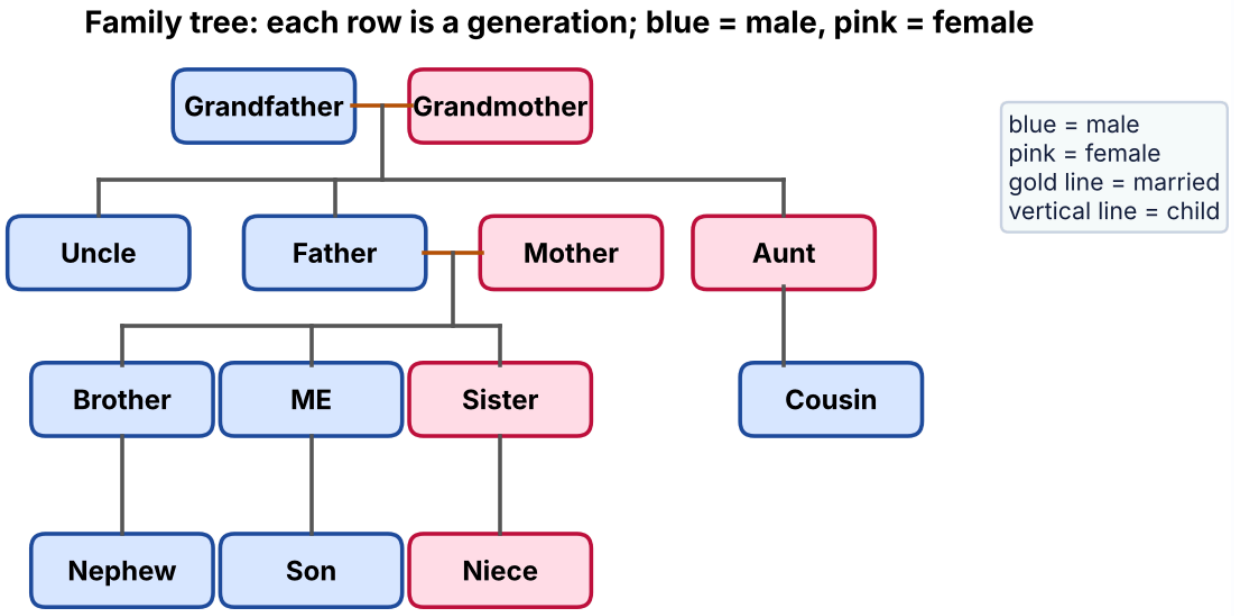
1. If DRINK is written as FTKPM, how is HOUSE written? — Basic
2. If CAT = 24 (sum of positions), find DOG. — Basic
3. If SPEED is coded as TQFFE, how is MOON coded? — Basic
4. If every letter is replaced by its opposite letter, code TRAIN. — Medium
5. FACE → HCEG. How is BOMB coded? — Basic
6. In a code, sun = ka, moon = mi, star = ro; (sun moon) = ka mi and (moon star) = mi ro. Code of star? — Medium
7. Using the shifts +1, +2, +3 ..., code MIND. — Medium
8. In a digit code ROSE = 4321 and BIRD = 5678. What is the code of the letter E? — Advanced

03 REASONING ABILITY · PART A · MUST-DO
Blood Relations

Blood relation questions describe a family in words. The method that never fails is: **draw a small family tree** with generations as rows, males as squares or blue boxes and females as circles or pink boxes. Do not try to solve it in your head.

Relation vocabulary

Relation	Meaning
Paternal grandfather / grandmother	father's father / father's mother
Maternal grandfather / grandmother	mother's father / mother's mother
Paternal uncle / aunt	father's brother / father's sister
Maternal uncle / aunt	mother's brother / mother's sister
Cousin	child of one's uncle or aunt
Nephew / niece	son / daughter of one's brother or sister
Brother-in-law	spouse's brother, sister's husband, spouse's sister's husband
Sister-in-law	spouse's sister, brother's wife
Father-in-law / mother-in-law	parents of one's spouse
Son-in-law / daughter-in-law	husband of daughter / wife of son
Co-brother / co-sister	husbands' brothers / wives' sisters (in some questions)



A general family tree. Use rows for generations and keep males and females visibly different.

Rules that decide most questions

- **Gender** is known only from words such as father, mother, brother, sister, son, daughter, husband, wife. “Cousin”, “parent”, “child”, “sibling”, “spouse”, “friend” do not tell the gender.
- **Generation**: parents are one row above, children one row below, siblings, spouses and cousins are in the same row.
- “My father’s only son” is the speaker (if the speaker is male).
- “My mother’s only daughter” is the speaker (if the speaker is female).
- “Son of my father’s sister” = cousin (maternal-side relationship is from the father’s side: **paternal aunt’s son**).
- Brother of mother = maternal uncle; sister of father = paternal aunt.
- If the gender of the person asked about is not given, the answer is the gender-neutral form (cousin, nephew/niece, sibling).

Pointing and puzzle statements

Example. Pointing to a girl, Meera said, “She is the daughter of the wife of the only son of my mother’s father.” My mother’s father’s only son is Meera’s maternal uncle; his wife is Meera’s maternal aunt; her daughter is Meera’s **cousin**.

Symbol based (coded) relations

Symbol coding: translate each symbol, then walk along the chain

P ★ Q	P ◆ Q	P ▲ Q	P ● Q
P is mother of Q	P is brother of Q	P is husband of Q	P is daughter of Q

P ◆ Q ★ R ▲ S → P is brother of Q, Q is mother of R, R is husband of S → P is maternal uncle of S's husband

Convert the symbols one at a time. Then travel along the chain from the first person to the last.

A is the brother of B. B is the sister of C. D is the father of A. How is C related to D?

Son or daughter

A, B and C are siblings (children of D). C’s gender is not given, so C is the son or daughter of D.

Pointing to a woman, Ravi says “She is the mother of the daughter of my wife.” How is the woman related to Ravi?

Wife

The daughter of Ravi’s wife is Ravi’s daughter; her mother is Ravi’s wife.

P is the mother of Q. R is the husband of S. S is the daughter of P. How is R related to Q?

Brother-in-law

S is P's daughter and so Q's sister (Q is P's child). R is S's husband, so he is Q's sister's husband = brother-in-law.

If A × B means A is the father of B and A + B means A is the sister of B, then in P + Q × R how is P related to R?

Paternal aunt

P is the sister of Q and Q is the father of R, so P is R's father's sister = paternal aunt.

Shortcuts and tricks

- Use the first-person word "me" as the centre of the tree and place everyone around it.
- Mark the generation difference first; then mark gender.
- When two people are called "brother-in-law", ask whose husband or brother they are.
- For puzzle-type family questions (7–8 persons), make a table of name, gender, generation and spouse.

Common mistakes

- Assuming gender from the name (never do it).
- Mixing paternal and maternal sides. Paternal = father's side. Maternal = mother's side.
- Forgetting that "only" closes the family (only son = one son).

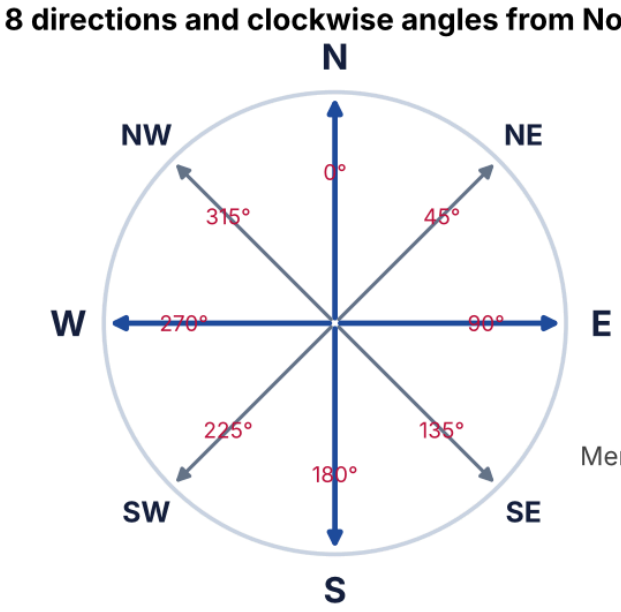
Practice set: Blood relation practice

1. A is the father of B. B is the sister of C. D is the mother of A. How is D related to C? — Basic
2. Pointing to a boy, Neha says "He is the son of my mother's only brother." Relation? — Medium
3. A and B are brothers. C is B's son. D is C's sister. How is A related to D? — Medium
4. P ★ Q means P is mother of Q, P ◆ Q means P is brother of Q. In P ◆ Q ★ R, how is P related to R? — Medium
5. If P is the sister of Q, Q is the sister of R and R is the husband of S, then P is S's ...? — Medium
6. Pointing to a photograph, a man says: "The man in the photograph is the son of my father's only son." How is the man in the photograph related to the speaker? — Advanced
7. A is the mother of B. B is the sister of C. C is the father of D. D is the daughter of E, who is C's wife. How is A related to E? — Advanced

REASONING ABILITY · PART A · IMPORTANT

04 Directions and Distances

A direction question follows a person walking, turning and walking again. Draw the path on a small grid with **North up, East right**. Every turn is a 90° rotation; at the end you either find the final **direction** the person faces, the **distance from the start**, or the **direction of one point from another**.



90° turns

Facing N	right → E	left → W
Facing E	right → S	left → N
Facing S	right → W	left → E
Facing W	right → N	left → S

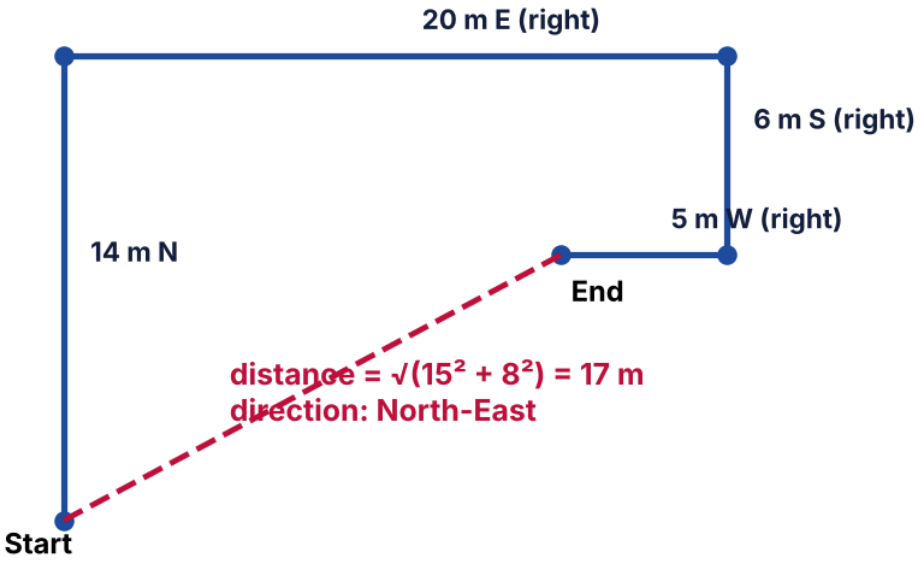
Memory trick: facing North, right = East and left = West. Turn the compass as you turn.

Learn the 8 directions and the effect of a right or left turn. Turn right = clockwise.

Formulas and rules

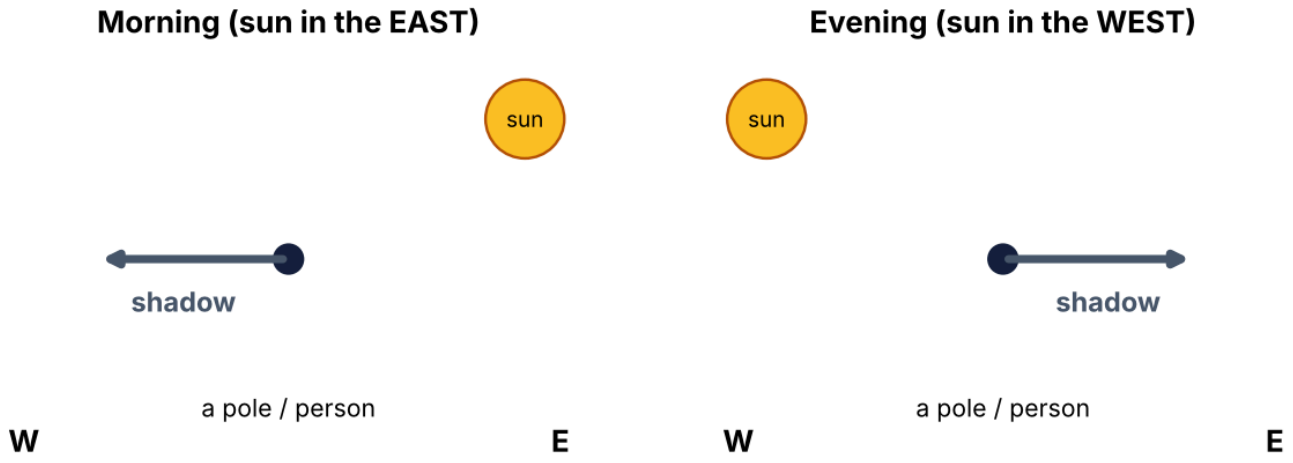
- Distance from start (straight line) = $\sqrt{(x^2 + y^2)}$ where x = net east–west, y = net north–south.
- Right turn = clockwise; left turn = anticlockwise. Facing North: right = East, left = West.
- Angles are clockwise from North: N 0°, NE 45°, E 90°, SE 135°, S 180°, SW 225°, W 270°, NW 315°.
- Pythagorean triples to remember: **3-4-5, 5-12-13, 8-15-17, 7-24-25, 20-21-29, 9-40-41** (and their multiples).
- Direction of B from A: compare coordinates. If B is east and north of A, then B is North-East of A.
- Shadow rule: **morning** shadow points West (sun in the East); **evening** shadow points East.
- At noon the shadow falls directly towards the North (in India).

Net displacement: add the east-west moves, add the north-south moves, then Pythagoras



Sample route. East–west total = +20 − 5 = 15 m. North–south total = 14 − 6 = 8 m. 15-8-17 is a Pythagorean triple.

Shadow questions



Morning: shadow falls to the west. If the shadow falls to your right in the morning, west is on your right, so you face South.

Ravi faces East. He turns 135° clockwise, then 180° anticlockwise, then 90° clockwise. Direction now?

South-East

East = 90°. +135° = 225° (SW). −180° = 45° (NE). +90° = 135° (SE).

A man walks 14 m North, turns right and walks 20 m, turns right and walks 6 m, turns right and walks 5 m. How far is he from the start and in which direction?

17 m, North-East

North 14 − South 6 = 8 m north; East 20 − West 5 = 15 m east; √(15² + 8²) = 17; the end point is north and east of start.

Just before sunset Mohan stands with his shadow falling exactly to his left. Which direction is he facing?

South

At sunset the shadow points East. East is on his left, so he faces South (facing South, left = East).

Raju goes 17 m North then 6 m East. Sanju goes 18 m West then 10 m North from the same point. Find the distance between them.

25 m

Raju is at (6, 17), Sanju at (−18, 10). Difference = (24, 7). $\sqrt{(24^2 + 7^2)} = 25$.

Shortcuts and tricks

- Draw a plus sign and write N, E, S, W before you start; keep the compass on your rough sheet.
- Do not use the word “left” on the page; use N, E, S, W for the person’s moves after you convert each turn.
- The straight-line distance is never the sum of the moves, only of the net x and net y.
- A net x of 0 or a net y of 0 means the person is exactly on a line; the direction is then only N, S, E or W.

Common mistakes

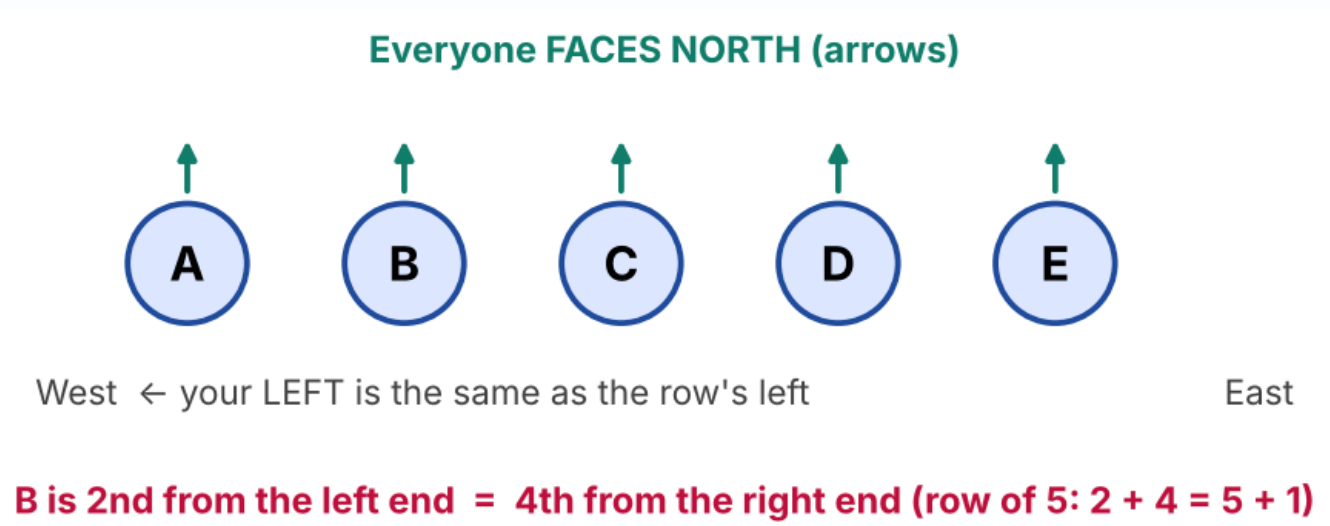
- Treating a left turn as a right turn when the person is facing South (facing South, left = East).
- Adding east and west moves instead of subtracting.
- Giving the direction of A from B when the question asks the direction of B from A (they are opposite).

Practice set: Direction practice

1.	A man goes 12 m North and 5 m East. How far is he from the start?	Basic
2.	Facing West, Sita turns 90° clockwise, then 90° clockwise again. Direction now?	Basic
3.	A person goes 3 km East, 4 km North, 3 km West. Distance from start?	Medium
4.	A goes 9 m West, turns right and goes 12 m. Distance from start?	Medium
5.	Facing the sun in the evening, in which direction is your right hand?	Medium
6.	P walks 40 m South, turns left and walks 9 m. Distance from start? Direction of start from P?	Advanced
7.	A man walks 8 m North, 6 m East, 8 m South and 3 m West. Distance from start and direction?	Advanced

05 REASONING ABILITY · PART A · VERY IMPORTANT
Seating Arrangements

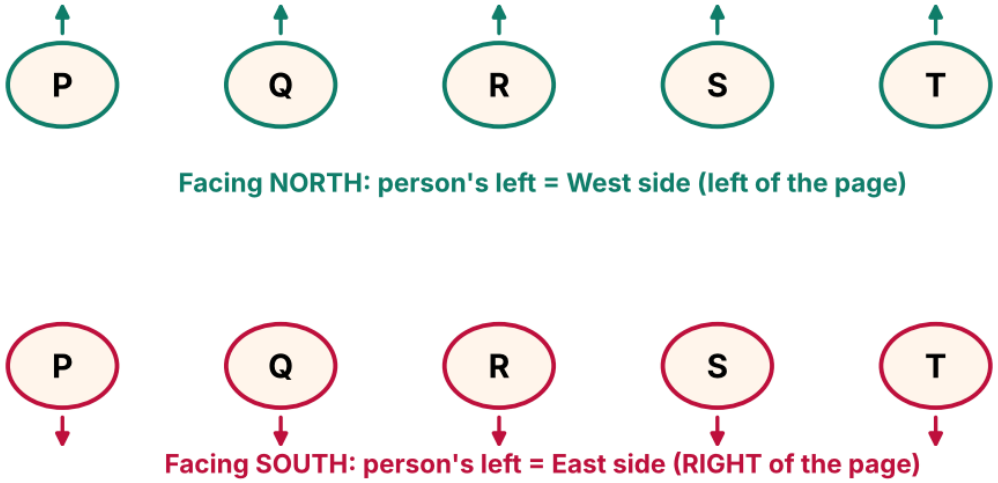
Seating arrangement is the **most repeated topic** in TCS reasoning. You are told who sits where relative to others; your job is to build a single consistent picture. Three layouts occur: **linear row**, **circle** and **two parallel rows**. The golden rule: **first find which way each person faces**, because left and right depend on it.



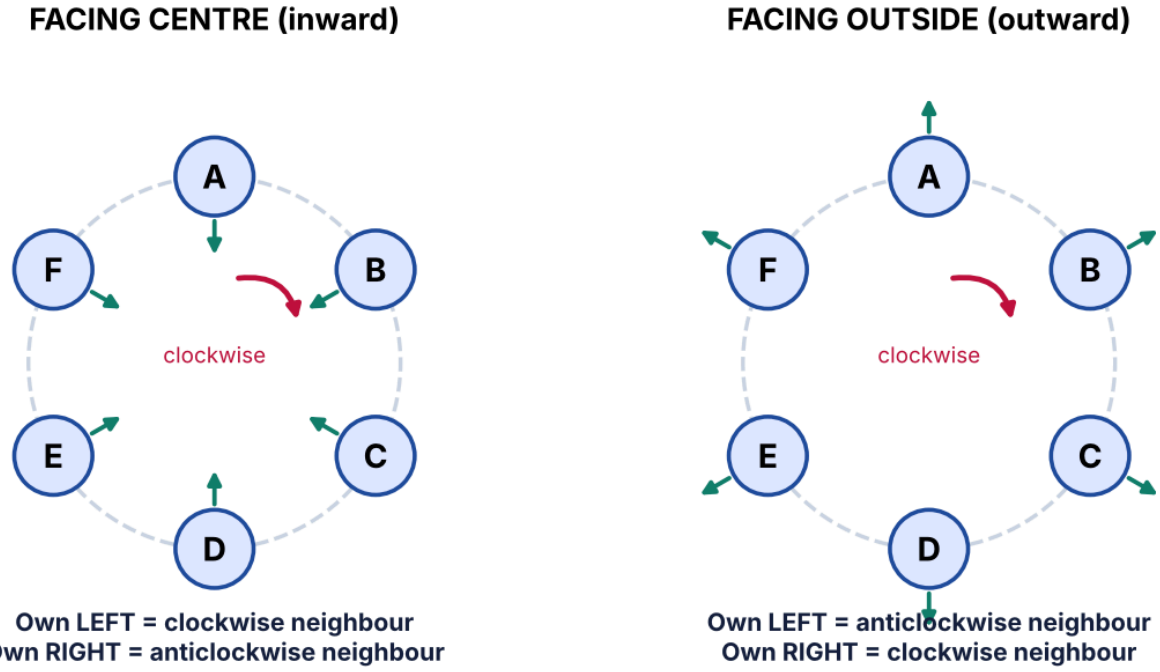
Linear arrangement: write positions 1 to n and fill the boxes. Left-end position + right-end position of the same person = n + 1.

Formulas and rules

- Position from left + position from right = **n + 1** (n = total persons).
- Persons **between** positions a and b (a < b) = **b – a – 1**.
- Facing North: your left = West, your right = East. Facing South it is the opposite.
- Circle with n persons: opposite person of position k is k + n/2 (n even). Persons between two people on the shorter arc = gap − 1.
- Circle: facing the centre, your **left = clockwise** neighbour; facing outside, your left = anticlockwise neighbour.
- Number of ways to seat n people: line n!, circle (n − 1)!, circle with bracelet symmetry (n − 1)!/2.
- Two rows: row 1 faces South and row 2 faces North; the person in front of you is the one directly across.



Facing direction decides left and right. When a question mixes “facing North” and “facing South” people, convert everything to compass directions.



Circular arrangement. The red arrow is clockwise. Inward-facing: left hand points clockwise. Outward-facing: left hand points anticlockwise.

Step-by-step method

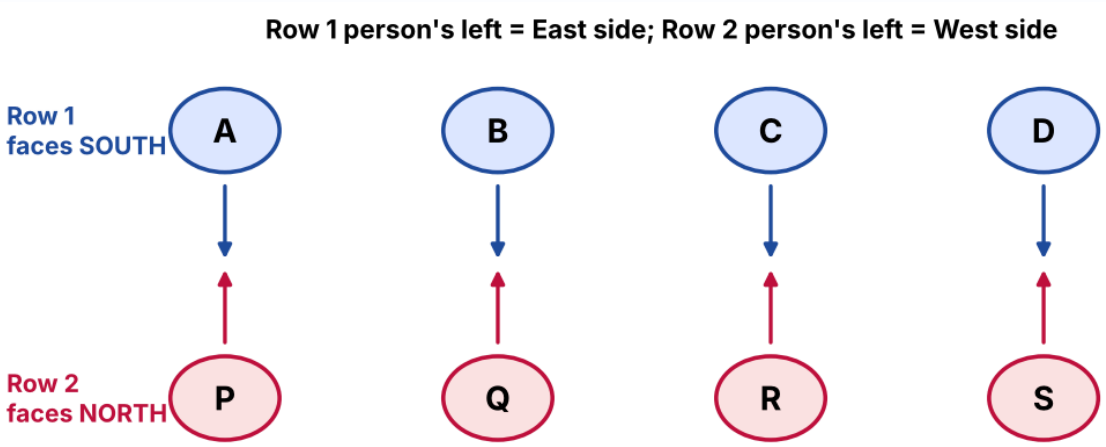
Step 1. Draw the shape (row, circle or two rows) and mark faces with arrows.

Step 2. Pick the statement with a **fixed position** (an end, “exactly in the middle”, “opposite to ...”) and place it first.

Step 3. Combine statements that name the same person into a chain (for example “A is left of B” and “B is left of C” gives A B C).

Step 4. If two arrangements are possible, keep both, and let the later statements eliminate one. Do not guess.

Step 5. Re-read each statement against the final picture once before answering.



Two-row (face-to-face) arrangement. Left/right of one row is the reverse of the other.

Six persons P, Q, R, S, T, U sit in a row facing North. Q is 2nd from the left. R sits immediately to the right of Q. S is at one of the ends. T sits immediately to the left of U. P is to the right of R and also sits at an end. Find the order from the left.

S Q R T U P

Q = position 2, R = position 3. S is at an end: position 1 or 6. T U must be adjacent (T then U) and P is right of R. S and P occupy the two ends; P is right of R so P is position 6 and S is position 1. T U fill positions 4, 5. Computer check (all 720 arrangements): the answer is unique: **S Q R T U P**.

Six friends A, B, C, D, E, F sit around a circle facing the centre. B is second to the left of A. C sits opposite A. D is immediately to the right of A. E is immediately to the left of C. Who sits between A and B?

F

Fix A at the top and count clockwise (left = clockwise for inward-facing people). Seats 0 to 5: A = 0, B = 2, C = 3, D = 5 (right = anticlockwise), E = 4, so F = 1. Clockwise order: A F B C E D. F sits between A and B.

Eight people A, B, C, D, E, F, G, H sit around a circle facing the centre. How many persons sit between A and B if B is the 3rd to the right of A?

2

3rd to the right (anticlockwise) means 3 seats away. Seats between = 3 − 1 = 2 on that side (the other side has 8 − 3 − 1 = 4).

Five boys A, B, C, D, E sit in a row facing South; from west to east they are in the order A B C D E. Five girls P, Q, R, S, T sit in the row opposite facing North; from EAST to WEST they are in the order P Q R S T. Who faces C?

R

Convert both rows to the same compass end. Girls from west to east: T S R Q P. C is 3rd from the west, so the girl 3rd from the west, R, faces C.

Shortcuts and tricks

- Facing partners in two rows have the same compass-end position.
- Use “immediate” = adjacent. “Between” always excludes the two named persons.
- In a circle, with n even, opposite pairs are n/2 apart. Use this to place the last person.
- If a statement is negative (“A is not next to B”), use it last to eliminate, never to place.

Common mistakes

- Mixing up the left of a person facing South with the left of the page.
- Counting “between” with both ends included.
- Forgetting that in a circle the seat after the last seat is the first seat (positions wrap).

Practice set: Seating practice

1. In a row of 25, A is 11th from the left. What is his position from the right? — Basic
2. In a row of 40 people, A is 14th from the left and B is 19th from the right. How many are between them? — Basic
3. 8 people sit in a circle. How many are between A and the person opposite A on either side? — Medium
4. P is 7th from the left and 5th from the right in a row. How many people are there? — Medium
5. In how many ways can 6 people sit around a circular table? — Medium
6. Eight friends sit around a circle facing the centre. B is 3rd to the left of A. C is exactly opposite to A. How many persons sit between B and C (shorter arc)? — Advanced
7. In a row of 5 facing North, P is immediately left of Q, R is at the right end, S sits between Q and R, T is not at the left end. How many valid orders exist? — Advanced

06 REASONING ABILITY · PART A · HIGH WEIGHT
Logical Puzzles and Arrangements

A puzzle gives a set of persons, a few attributes (floor, day, pet, city, colour) and a list of clues. Exactly **one** consistent arrangement exists. The skill is not guessing, it is **building a table** and filling it in the right order: **fixed information first, negatives last**.

Formulas and facts used in puzzle questions

- Knockout tournament with n teams: matches = **n − 1**; rounds = $\log_2 n$ (n = power of 2).
- Round robin (each pair once): matches = **n(n − 1)/2**; twice (home and away): n(n − 1).
- Number of handshakes among n people = n(n − 1)/2.
- Days of the week cycle with 7: "the day after the day after tomorrow" = +2 + 1 = 3 days later.
- If k people sit in one row of m, number of persons between positions a and b = |a − b| − 1.
- Comparison puzzles (taller, older, heavier): a chain A > B > C lets you place any element by position; use an inequality line.

1. Floor / order puzzles

Floor 6	C
Floor 5	E
Floor 4	D
Floor 3	A
Floor 2	F
Floor 1	B

Clues (6 persons, 6 floors):

1. C lives on the top floor (6).
2. A lives two floors above B.
3. E lives directly above D.
4. F lives between B and A.
5. E lives above A.

Solve: C = 6 → fixed. F is between B and A, A = B + 2, so B, F, A are consecutive: (1,2,3), (2,3,4) or (3,4,5). E above A and E directly above D leaves only B = 1:
B1, F2, A3, D4, E5, C6.

Draw the building. Place the person with a fixed floor first, then use "between", "directly above" and "two floors above" to connect the others.

Six persons A–F live on six different floors (1 = lowest). C lives on the top floor. A lives two floors above B. E lives directly above D. F lives between B and A. E lives above A. Who lives on floor 4?

D

B, F, A are consecutive floors (F is the middle). E is directly above D and above A, so the pair D–E sits above the trio B-F-A. The only fit is B1, F2, A3, D4, E5, C6. Checked by trying all 720 arrangements: exactly one works.

2. Table / grid puzzles

	Tea	Coffee	Milk	Cat	Dog	Fish
Amit	×	✓	×	✓	×	×
Bina	×	×	✓	×	×	✓
Chetan	✓	×	×	×	✓	×

Fill a ✓ and cross out its row and column. Final grid above.

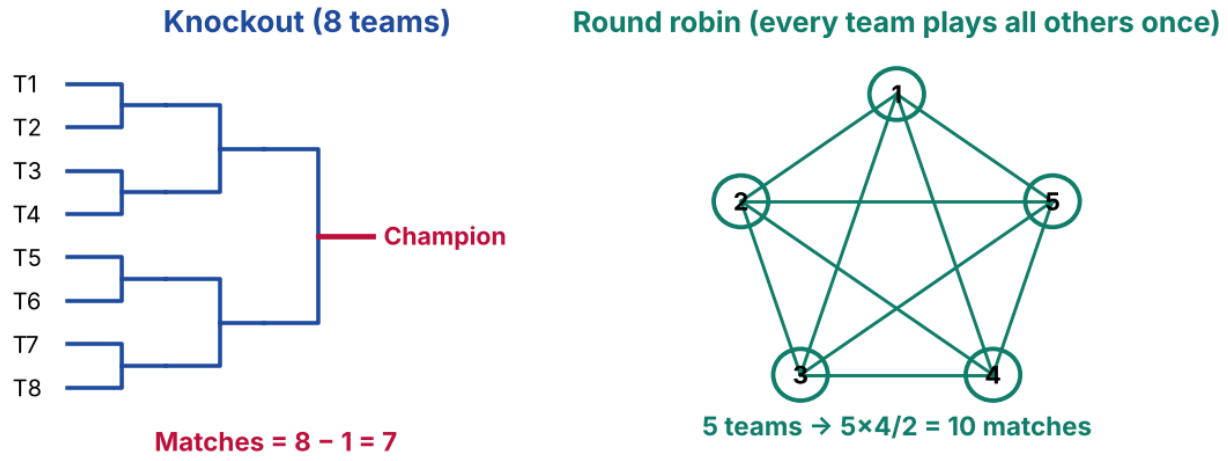
Use a yes/no matrix. When you put a tick, the row and column of that tick become crosses automatically.

Amit, Bina and Chetan each like one drink (Tea, Coffee, Milk) and keep one pet (Cat, Dog, Fish). Amit likes neither Tea nor Milk. Bina keeps no Dog and does not drink Coffee. The Coffee drinker keeps the Cat. Chetan keeps no Fish and does not drink Milk. Who keeps the Fish and what does that person drink?

Bina; Milk

Amit has Coffee (not Tea, not Milk). Coffee drinker has the Cat, so Amit has the Cat. Chetan does not drink Milk, so Chetan has Tea and Bina has Milk. Chetan has no fish, so Chetan has the Dog, leaving Fish for Bina. Unique solution confirmed by listing every 36 combinations.

3. Tournament / counting puzzles



Knockout: every match removes exactly one team, so n − 1 matches crown the winner. Round robin: choose any two teams, C(n, 2).

In a knockout tournament 32 teams take part. How many matches are played and how many rounds?

31 matches, 5 rounds

Each match eliminates exactly one team. To leave one champion, 31 teams must be eliminated: 31 matches. 32 → 16 → 8 → 4 → 2 → 1 is 5 rounds.

Twelve friends meet and each shakes hands once with every other. How many handshakes?

66

C(12, 2) = 12 × 11 / 2 = 66.

4. Days and comparison puzzles

Five boys are compared in height. A is taller than B but shorter than C. D is taller than C but shorter than E. Who is the shortest and who is the tallest?

B shortest, E tallest

Chain: E > D > C > A > B. The first in the chain is tallest, the last is shortest.

Shortcuts and tricks

- Start with the clue that gives a **definite place**; leave "not" clues until the end.
- If a clue links three people, write them as a chain and slide the chain on the table.
- Use ✓ / × everywhere; a column with only one cell left must be a tick.
- Re-read the question; many puzzles ask for the person next to or two floors below, not the main attribute.
- Time target: 2 to 3 minutes for a 5 to 6 person puzzle.

Common mistakes

- Writing a guess into the table. Only forced facts go in.
- Forgetting that "between" means strictly between.
- Using "higher than" as "immediately above".

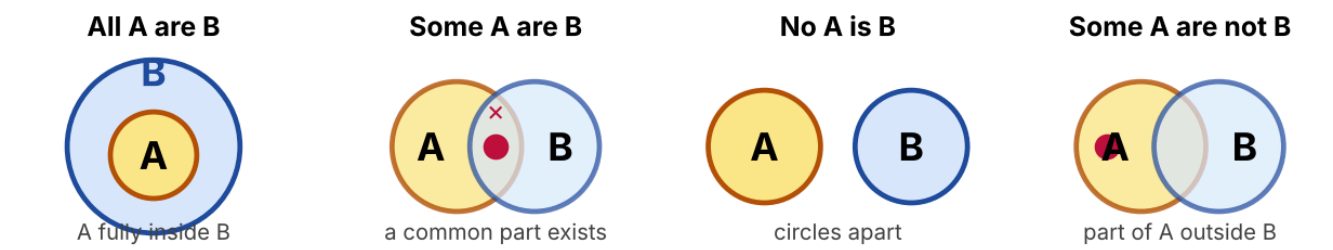
Practice set: Puzzle practice

1.	In a knockout tournament with 64 teams, how many matches in total?	Basic
2.	In a round robin with 9 teams, how many matches are played?	Basic
3.	A > B, C > A, D < B. Who is the smallest and who is the largest?	Medium
4.	In a league each of 8 teams plays every other twice. Total matches?	Medium
5.	Six people stand in a line. P is 4th from the left and Q is 2nd from the right. How many stand between them?	Medium
6.	Four friends A, B, C, D have different ages. A is older than C. B is younger than D but older than A. Who is the oldest, who is the youngest?	Advanced
7.	Ten people are in a room. Everyone shakes hands with everyone once, except 3 particular people who refuse to shake hands with each other. Handshakes?	Advanced

07 Syllogism

REASONING ABILITY · PART A · VERY IMPORTANT

A syllogism gives two or three statements and asks which conclusions **definitely follow**. Treat the statements as **true even when they sound silly** (“All pens are roads”). Use Venn diagrams, or the quick rules below. TCS also asks “possibility” conclusions, which need a second look.

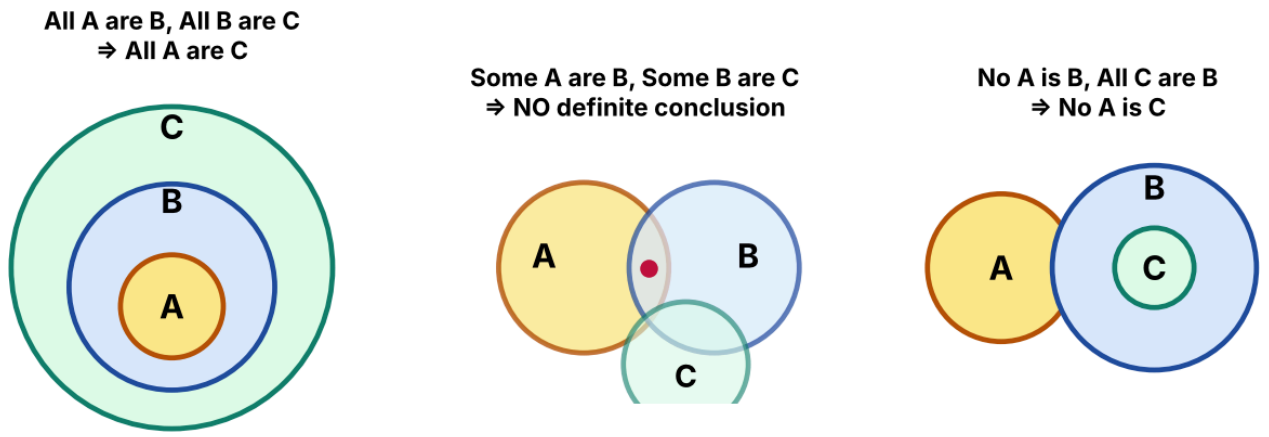


The four basic statements. Draw them once and the rest of the topic becomes easy.

Statement	Name	Convert (valid)	Cannot convert
All A are B	Universal affirmative (A)	Some B are A	All B are A
Some A are B	Particular affirmative (I)	Some B are A	All A are B
No A is B	Universal negative (E)	No B is A	—
Some A are not B	Particular negative (O)	nothing (no valid conversion)	Some B are not A

Rules that follow from the diagrams

- **All + All → All**: All A are B, All B are C ⇒ All A are C.
- **All + No → No**: All A are B, No B is C ⇒ No A is C.
- **Some + All → Some**: Some A are B, All B are C ⇒ Some A are C.
- **All + Some → no definite conclusion** (middle term not guaranteed to overlap).
- **Some + Some → no definite conclusion**; **No + No → no conclusion**; **Some not + anything → check by diagram**.
- Conversion: “All A are B” gives “Some B are A”; “Some A are B” gives “Some B are A”; “No A is B” gives “No B is A”.
- **Either–or case**: if two conclusions are a complementary pair (for example “Some A are B” / “No A is B”) and neither alone follows, then “either I or II follows”.
- Possibility: a conclusion “Some A can be B” is correct if the diagram can be drawn so that A and B overlap without breaking the statements.



Three standard chains drawn with Venn diagrams. The red dot is a definite member of the overlap.

Method: the 3-step Venn routine

Step 1. Draw each statement separately, then merge them into the **most spread-out** picture consistent with all statements (circles overlap only when forced).

Step 2. A conclusion is **definite** only if it is true in **every** picture you can draw.

Step 3. A conclusion with the word "possible" is acceptable if it is true in **at least one** picture.

Statements: All books are pens. All pens are keys. Conclusions: I. All books are keys. II. Some keys are books. Which follow?

Both I and II

All + All gives "All books are keys" (I). Converting gives "Some keys are books" (II). Both follow. (Verified with an exhaustive model check over all 256 region models.)

Statements: Some cats are dogs. All dogs are rats. Conclusions: I. Some cats are rats. II. Some rats are cats. III. All cats are rats. Which follow?

I and II only

Some + All gives "Some cats are rats", and its converse. III would need every cat to be a dog, which is not given.

Statements: All A are B. No B is C. Conclusions: I. No A is C. II. No C is A. III. Some A are C.

I and II follow

Since A is inside B and B does not meet C, A does not meet C (I), and the converse of a "No" statement also holds (II). III contradicts the statements.

Statements: Some A are B. Some B are C. Conclusions: I. Some A are C. II. No A is C. Which follow?

Either I or II

Neither alone is definite: the A–C overlap may exist or not. But one of the two must be true, because they are complementary. So the answer is "Either I or II follows". (Checked: both conclusions are possible and not forced.)

Shortcuts and tricks

- Draw circles only for what is stated; never add an overlap that is not forced.
- If a conclusion uses “All” and the statements only have “Some”, it almost never follows.
- A pair like “Some X are Y” and “No X is Y” is the classic either–or pair; “All X are Y” and “Some X are not Y” as well.
- Statements are facts, not your general knowledge. Ignore what you know about the real world.
- A “No” statement together with an “All” can only give a “No” conclusion, never a positive “Some” conclusion. (Textbooks differ on whether “No A is B” also gives “Some A are not B”; TCS questions avoid depending on it.)

Common mistakes

- Concluding “All A are C” from “Some A are B, All B are C”.
- Using real-world knowledge to accept or reject a conclusion.
- Forgetting to test the converse of every conclusion before writing “only I follows”.

Practice set: Syllogism practice

1. All roses are flowers. Some flowers are red. Conclusion: Some roses are red. Follows? — Basic

2. Some men are doctors. All doctors are teachers. Conclusion: Some men are teachers. Follows? — Basic

3. All A are B. All C are B. Conclusion: Some A are C. Follows? — Medium

4. No pen is a book. All bags are books. Conclusion: No bag is a pen. Follows? — Medium

5. All trees are plants. No plant is stone. Conclusion: No tree is a stone. Follows? — Medium

6. Some A are B. No B is C. Conclusion: Some A are not C. Follows? — Advanced

7. All A are B. Some B are C. Conclusions: I. Some A are C. II. No A is C. Which follows? — Advanced

Clock and Calendar

These two topics are pure formula. They give quick marks if you learn **five clock formulas** and the **odd-days method**. A clock face is 360°, the minute hand moves 6° per minute, the hour hand 0.5° per minute, so the minute hand gains 5.5° every minute.



Three standard clock situations.

Clock formulas

- Angle between hands at H hours M minutes = **|30H − 5.5M|** (if more than 180°, subtract from 360°).
- Hour hand: 30° per hour = 0.5° per minute. Minute hand: 6° per minute. Relative speed 5.5° per minute.
- Hands overlap **11 times** in 12 hours (22 times in a day). Overlap after H o'clock: **t = 60H/11** minutes past H.
- Hands are at right angles **22 times** in 12 hours (44 in a day); in a straight line (180°) **11 times** in 12 hours.
- Mirror image of a time: **11:60 − T** (or 12:00 − T when the time is exactly on the hour).
- A clock that gains or loses x minutes per hour shows correct time again after 12×60/x hours of the correct clock.

Find the angle between the hands at 3:40.

130°

|30×3 − 5.5×40| = |90 − 220| = 130°. Less than 180°, so the answer stays 130°.

At what time between 2 and 3 do the hands overlap?

2:10 10/11

t = 60H/11 = 120/11 = 10 10/11 minutes after 2.

What is the mirror image of 4:25?

7:35

11:60 − 4:25 = 7:35.



Odd days and the century table.

Calendar rules

- Leap year: divisible by 4; century years only if divisible by **400** (1900 is not a leap year, 2000 is).
- Odd days: ordinary year 1, leap year 2. Months: Jan 3, Feb 0 (1 in leap), Mar 3, Apr 2, May 3, Jun 2, Jul 3, Aug 3, Sep 2, Oct 3, Nov 2, Dec 3 (31-day month: 3 odd days; 30-day: 2).
- Day-count: Sun = 0, Mon = 1, ... Sat = 6. Total odd days mod 7 gives the weekday.
- Two years have the same calendar if they start on the same weekday and are both leap or both ordinary (for example 2023 and 2034 repeat; 2026 and 2037 repeat).
- Number of Sundays in a 30-day month starting on Sunday = 5; the day after 100 days = 100 mod 7 = 2 days later.

On what day of the week was 15 August 1947?

Friday

1946 years = 1600 (0 odd) + 300 (1 odd) + 46 years (46 + 11 leap years = 57, 57 mod 7 = 1) = 2 odd days. In 1947: Jan–Jul = 31+28+31+30+31+30+31 = 212 days, plus 15 = 227 days. Total odd days = (2 + 227) mod 7 = 5, i.e. Friday (0 = Sunday). (Checked with the Python calendar: Friday.)

26 January 1950 was a Thursday. What day was 26 January 1952?

Saturday

1950 → 1951 adds 1 odd day, 1951 → 1952 adds 1 odd day, but the Feb 29, 1952 is after 26 Jan so is not counted. Thursday + 2 = Saturday. (Checked by calendar.)

Shortcuts and tricks

- For the days from a known date: count the number of days, divide by 7 and add the remainder to the known weekday.
- When the date lies **before** 29 Feb in a leap year, do not count the leap day for that year.
- For a mirror image use 11:60 − T. A time that is exactly on the hour, such as 4:00, is subtracted from 12:00, giving 8:00.
- The first two clock formulas answer 80% of all clock questions.

Common mistakes

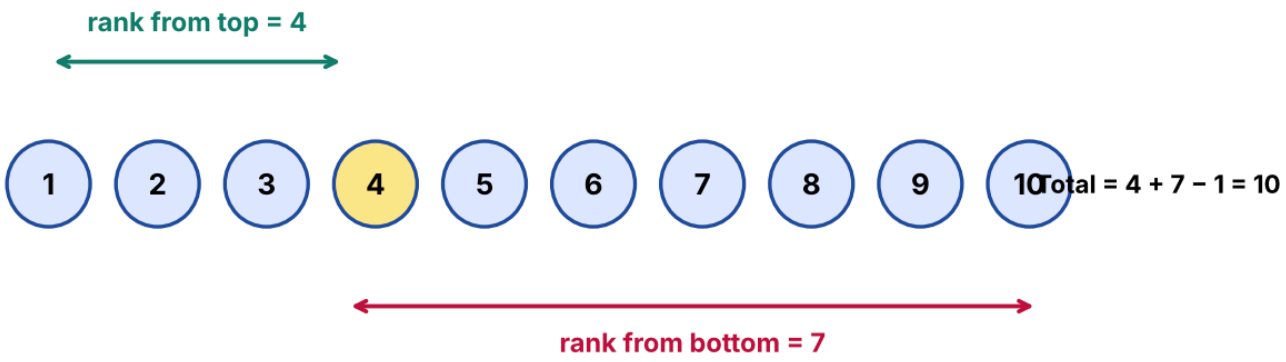
- Using 30H − 5.5M and forgetting to reduce angles above 180°.
- Treating 1900, 2100 as leap years.
- Counting the start date and the end date both when finding days between two dates.

Practice set: Clock and calendar practice

1.	Angle between the hands at 6:30?	Basic
2.	Angle between the hands at 9:15?	Basic
3.	How many times do the hands of a clock overlap in 24 hours?	Medium
4.	Mirror image of 8:20?	Medium
5.	If 1 January 2026 is a Thursday, what day is 1 January 2027?	Medium
6.	How many odd days are there in 400 years?	Medium
7.	26 January 2026 falls on which day of the week?	Advanced

Ranking, Alphabet and Number Tests

These three topics are mostly counting. The same few ideas appear again and again: **position from the left plus position from the right equals total plus one**, the alphabet has 26 letters and its mirror positions add up to 27, and numbers can be searched for patterns by writing them out.



Rank from top + rank from bottom = total + 1.

Ranking formulas

- **Total = (rank from top) + (rank from bottom) − 1.**
- People between two persons with ranks a < b from the top = **b − a − 1.**
- If A is a-th from the left and B is b-th from the right in the same row and they exchange places: new position of A from the left = given position of B from the left, and so on. Use positions to track the exchange.
- If two rankings are given (class rank from top and bottom), total students = sum − 1.
- When A moves up k places in a ranking, the new rank = old rank − k. Persons in between lose one rank each.

Ravi is 14th from the top and 27th from the bottom of a class. How many students are there?

40

Total = 14 + 27 − 1 = 40.

In a row, A is 12th from the left and B is 15th from the right. After they exchange seats, A is 25th from the left. How many persons are in the row?

39

After the exchange A sits where B used to sit. So B's original seat is 25th from the left and 15th from the right. Total = 25 + 15 − 1 = 39.

Forward position (red) with vowels highlighted

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1

Backward position (green): letter N-th from the end

Forward position + Backward position of the SAME letter = 27. Opposite letter of position p is at 27 − p.

Memory: EJOTY = 5, 10, 15, 20, 25 · JLQ, ACG, BDF ... "Z-Y-X" reverse spelling trick

Alphabet table: forward and backward positions. Vowels are highlighted.

Alphabet tricks

- **EJOTY**: E = 5, J = 10, O = 15, T = 20, Y = 25.
- Mirror letter of a letter at position p is the letter at **27 − p** (A↔Z, B↔Y, M↔N).
- Letter-to-letter distance: $|p - q|$; letters between = $|p - q| - 1$.
- Alphabetical position of a word's letters can be summed: CAT = 3 + 1 + 20 = 24.
- Reverse of a word: write the letters backward; the n-th from the left is the n-th from the right in the reverse.
- Vowels: A E I O U. If a question names "the vowel immediately after the 4th letter", count first, then search forward.

If the letters of the word COMPUTER are arranged in alphabetical order, how many letters keep their position?

2

Alphabetical order: C E M O P R T U. Original: C O M P U T E R. Position 1: C = C ✓. Position 3: M = M ✓. The other six differ. Answer 2. (Checked by code.)

How many letters lie between H and S in the alphabet?

10

H = 8 and S = 19. Between = 19 − 8 − 1 = 10.

Number tests

In the series 4 3 5 7 3 5 3 2 5 3 5 2 3 5, how many 3s are immediately followed by 5 and not immediately preceded by 7?

3

The pattern "3 5" begins at positions 2, 5, 10 and 13. The 3 at position 5 is preceded by 7, so it is excluded. That leaves positions 2, 10 and 13: three. (Verified by code.)

Shortcuts and tricks

- Write the sequence with positions; do not count mentally.
- For “how many X preceded by A and followed by B” scan every X and test its two neighbours.
- For letter-gap questions draw a line of letters and count carefully.
- Memorise the position of A, E, J, O, T, Y, M, N, Z.

Common mistakes

- Using total = top + bottom without subtracting 1.
- Counting a person twice when finding persons between.
- Forgetting that A–Z mirror positions add to 27, not 26.

Practice set: Ranking and alphabet practice

1. Rahul ranks 7th from the top in a class of 35. What is his rank from the bottom? **Basic**

2. What is the 10th letter to the right of the 5th letter from the left end? **Basic**

3. The letter 4th to the right of the 12th letter from the right end of the alphabet is? **Medium**

4. In a class of 48, Seema is 17th from the top. Her friend Mira is 20 ranks below her. Mira’s rank from the bottom? **Medium**

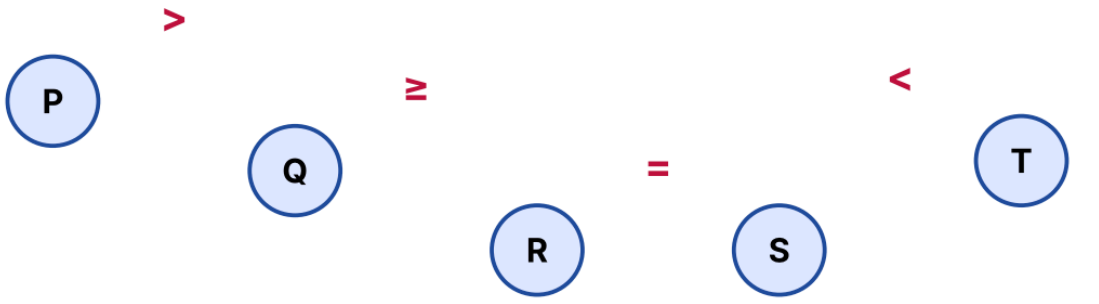
5. How many letters in the word PROBLEM keep their position when arranged alphabetically? **Medium**

6. How many pairs of letters in the word ALGORITHM have as many letters between them in the word as in the alphabet? **Advanced**

7. If A = 1, B = 2, ... then the value of REASON is? **Advanced**

10 REASONING ABILITY · PART A · QUICK SCORING
Inequalities and Symbol Operations

In the inequality chain you receive a chain of symbols and decide which conclusions are **definitely true**. Symbols can also be **coded** ("A @ B means A > B"). For mathematical operations the signs themselves are swapped and you evaluate normally with BODMAS.



P > Q ≥ R = S < T ⇒ P > R ✓, Q ≥ S ✓, T > R ✓ but Q > S ×, P ? T (no relation)

Combine only when the signs point the SAME way (both > / ≥, or both < / ≤).

A chain is read left to right. Only like-direction signs combine. Verified by testing every value 1 to 6.

Rules of combination

- > and ≥ combine to > (strict wins): P > Q ≥ R ⇒ P > R.
- ≥ and ≥ combine to ≥; = does not change the other sign: Q ≥ R = S ⇒ Q ≥ S.
- > with < (opposite directions) ⇒ **no conclusion** between the end terms: P > Q < R gives nothing about P and R.
- For "either ... or" pairs: when the two conclusions are (X > Y) and (X ≤ Y) or (X = Y) and (X ≠ Y), the pair is complementary and exactly one holds.
- A conclusion A ≥ B follows if the chain gives A > B **or** A = B **or** A ≥ B.
- Sign shortcuts: x ≥ y is "not x < y"; x ≠ y never follows from ≥.

Statement: P > Q ≥ R = S < T. Which of these definitely follow? (a) P > R (b) Q > S (c) T > R (d) Q ≥ S (e) P > T

(a), (c), (d)

(a) P > Q ≥ R gives P > R. (b) Q ≥ R = S gives only Q ≥ S, not >. (c) R = S < T gives T > R. (d) true. (e) P and T are linked only through opposite directions, no conclusion. (All five tested by program over every value 1 to 6.)

Coded inequalities

Symbols: A @ B means A > B; A # B means A ≥ B; A $ B means A < B; A % B means A = B.
Statement: P @ Q # R % S. Which is true: R ≤ Q, P > S, or both?

Both

P > Q ≥ R = S. So Q ≥ R (R ≤ Q ✓) and P > Q ≥ S gives P > S ✓.

Mathematical operations (sign swaps)

If + means ×, × means −, − means ÷ and ÷ means +, find 12 + 3 × 4 − 2 ÷ 5.

39

Replace the signs: 12 × 3 − 4 ÷ 2 + 5. BODMAS: 12×3 = 36; 4÷2 = 2; 36 − 2 + 5 = 39.

Operation task	Method
Signs swapped	Replace each sign by its new meaning first, then apply BODMAS.
Numbers swapped (e.g. 3 means 5)	Replace numbers first, then evaluate.
Find the missing sign (... 8 ? 4 ? 2 = 6)	Test the four signs in turn.
Balance an equation	Convert the right side to the left's value.

Shortcuts and tricks

- Write the full chain on the page and mark the direction; do not hold it in the head.
- Always compare the two end terms of the conclusion through the chain that joins them.
- A conclusion with ≥ is true when the chain gives > or = or ≥ between the same two terms.
- For sign-swap problems, rewrite the complete expression before calculating.

Common mistakes

- Combining > with <.
- Using = as if it were >.
- Evaluating the original sign when the question swapped it.

Practice set: Inequality practice

1. P > Q = R ≥ S. Does P > S follow? — Basic
2. A ≥ B > C < D. Does A > D follow? — Basic
3. If + means −, − means ×, × means ÷, ÷ means +, find 18 + 4 − 3 × 6. — Medium
4. A ≥ B ≥ C = D ≥ E. Does D ≥ E and A ≥ E follow? — Medium
5. If > means +, < means −, find 9 > 6 < 3 (the result). — Medium
6. P < Q ≤ R > S. Is P < R certain? Is Q > S certain? — Advanced
7. A = B ≥ C < D = E. Conclusions: I. A ≥ C II. D > C III. E ≥ B. Which follow? — Advanced

REASONING ABILITY · PART A · QUICK SCORING

11 Analogy, Classification and Odd-One-Out

Analogy asks you to find the **relationship** between the first pair and apply it to the second pair. Classification asks which item does not belong. Both are solved by naming the relationship in **one clear sentence** before looking at options.

Eight analogy relationships that cover most questions

Worker – Tool Carpenter : Saw	**Part – Whole** Petal : Flower	**Cause – Effect** Fire : Smoke	**Place – Product** Assam : Tea
Male – Female Lion : Lioness	**Degree** Warm : Hot	**Symbol** Dove : Peace	**Number rule** 8 : 64 :: 7 : 49

Eight relationships that cover most analogy questions.

Types and tests

- **Word analogy:** A : B :: C : ? → state the A–B relation (worker–tool, part–whole, cause–effect, place–product) and apply it.
- **Letter analogy:** use positions: AB : CD means +2 shifts; reverse and mirror pairs (A↔Z) are common.
- **Number analogy:** look for squares, cubes, ×/+ rules, digit sums. 5 : 125 :: 6 : 216 (cube).
- **Classification:** find the property shared by three of four items (all primes, all cubes, all even, all vowels...).
- **Number odd-one-out tests:** prime, perfect square/cube, digit sum, divisibility, alternate parity, ratio between digits.
- **Mixed groups:** in the group 2, 3, 5, 9, 11, 13, only 9 is composite; check each number against the likely property.

Dog : Kennel :: Bee : ?

Hive

Animal and its home. A dog lives in a kennel; a bee lives in a hive.

8 : 64 :: 11 : ?

121

Number is squared ($8^2 = 64$), so $11^2 = 121$.

5 : 125 :: 7 : ?

343

Cube: $5^3 = 125$, $7^3 = 343$.

CAT : DBU :: DOG : ?

EPH

Each letter moves one step forward: C→D, A→B, T→U; so D→E, O→P, G→H.

Find the odd one: 3, 5, 7, 9, 11

9

All others are prime; 9 = 3 × 3 is composite.

Find the odd one: 16, 25, 36, 48, 64

48

16, 25, 36 and 64 are perfect squares; 48 is not.

Shortcuts and tricks

- Say the relationship in a sentence: "A is the ... of B".
- Check the direction (A:B and C:D must be in the same order).
- For letter analogy, convert to numbers once and look at differences.
- Odd-one-out numbers: try prime, square, cube, divisibility by 3, 5, 7 in that order.

Common mistakes

- Choosing the option that merely "feels connected".
- Reversing the order of the relationship (Part:Whole vs Whole:Part).
- Ignoring plural/singular or tense in word analogies.

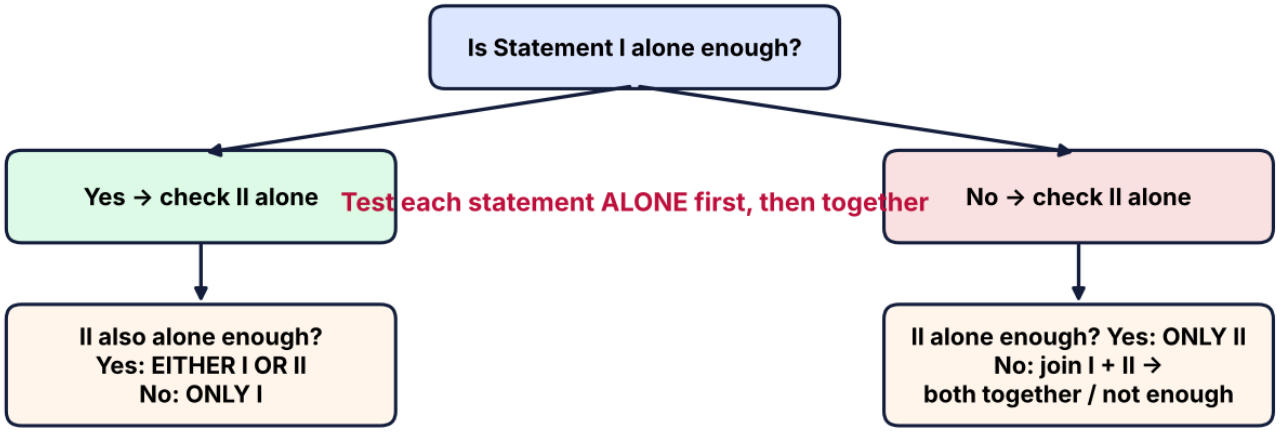
Practice set: Analogy and odd-one-out practice

1.	Pen : Write :: Knife : ?	Basic
2.	Book : Library :: Painting : ?	Basic
3.	3 : 27 :: 4 : ?	Medium
4.	AZ : BY :: CX : ?	Medium
5.	Find the odd one: 2, 3, 5, 7, 9	Medium
6.	ACE : GIK :: BDF : ?	Advanced
7.	Find the odd one: 144, 169, 196, 220, 256	Advanced

12 Data Sufficiency

REASONING ABILITY · PART A · TRICKY

A data sufficiency question has a question and two statements. You do **not** solve the question completely; you only decide whether the statements give enough information to solve it. The answer options are always the same five.



Fixed procedure. Do not skip the "alone" tests.

Option	Meaning
A	Statement I alone is sufficient; II alone is not.
B	Statement II alone is sufficient; I alone is not.
C	Both together are needed; neither alone is sufficient.
D	Each statement alone is sufficient (either one).
E	Even together they are not sufficient.

Principles

- "Sufficient" means a **single definite answer** (a unique value, or a definite Yes/No).
- For a value question, N unknowns need N independent equations; two equations x + y = 10 and 2x + 2y = 20 are not independent.
- For Yes/No questions, a statement is sufficient only if it gives **always Yes** or **always No**.
- Never combine the statements until each alone has been tested.
- Do not use outside knowledge unless it is a well-known fact (a week has 7 days).

What is the value of x? I. x + y = 10. II. x – y = 2.

Both together (C)

Alone, each has two unknowns. Together: x = 6, y = 4 (unique).

Is the integer N even? I. N + 3 is odd. II. N is divisible by 3.

Only I (A)

I: N + 3 odd means N is even, definite. II: multiples of 3 may be even or odd (6, 9), so not enough.

How many students are in the class? I. There are 12 girls. II. Girls are 40% of the class.

Both together (C)

12 = 0.4 × total gives 30.

On which day of the week was the meeting held? I. It was held 3 days after Sunday. II. It was held on the day before Thursday.

Either (D)

I: Sunday + 3 = Wednesday. II: the day before Thursday = Wednesday. Each alone suffices.

Shortcuts and tricks

- Make a mini-table of what each statement tells you.
- For ranking/seating data sufficiency, draw the arrangement using only that statement.
- Do not “help” the statement by assuming; if the unknown could take two values, it is insufficient.
- Check tricky Yes/No: a “Sometimes yes, sometimes no” is NOT sufficient.

Common mistakes

- Combining the statements first and answering C without testing alone.
- Thinking “I can solve it approximately” is enough.
- Treating “No” as insufficient: a definite “No” is sufficient.

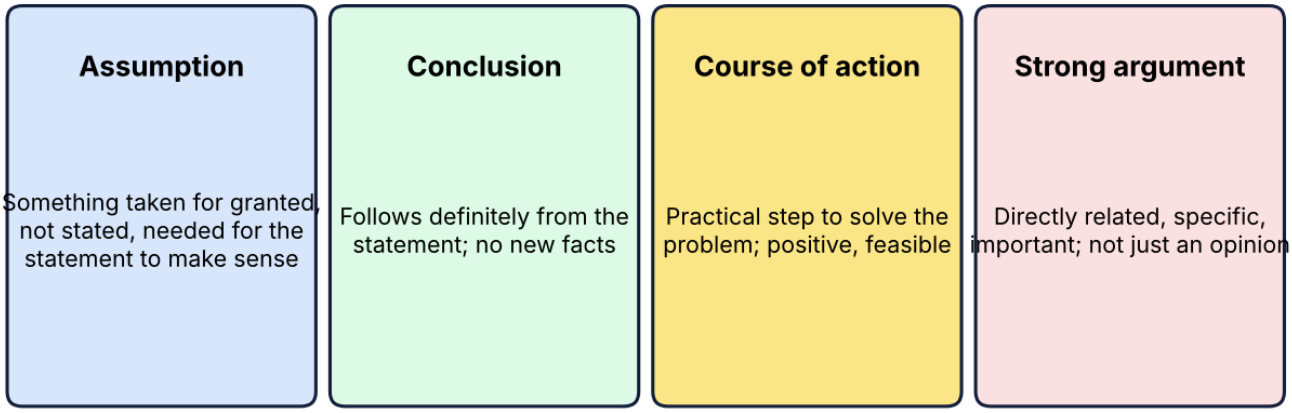
Practice set: Data sufficiency practice

1.	Is $x > 5$? I. $x^2 = 49$. II. $x > 0$.	Basic
2.	What is Raj’s age? I. Raj is 5 years older than Sam. II. Sam is 20.	Basic
3.	Is P the tallest? I. P is taller than Q. II. Q is taller than R.	Medium
4.	What is the area of a square? I. Perimeter is 20. II. Diagonal is $5\sqrt{2}$.	Medium
5.	Is N divisible by 6? I. N is divisible by 2. II. N is divisible by 3.	Medium
6.	Find the number: I. It is a two-digit prime. II. Its digits sum to 4.	Advanced
7.	Find x: I. $x^2 = 16$. II. $x > 0$.	Advanced

REASONING ABILITY · PART B · VERBAL LOGIC

13 Statement-Based Reasoning

These questions test **critical reading**, not mathematics. Four formats appear: **assumptions, conclusions, course of action, strong/weak arguments**, plus **cause and effect** and **decision making**. You must separate what is *stated* from what is *only suggested*.



The four core formats at a glance.

Format	What to ask	Accept when
Assumption	"Must this be true for the statement to hold?"	Implicit, necessary, not a restatement.
Conclusion	"Is this definitely supported?"	Follows from the statement alone; no extra data, no extreme words (all, only, never) unless stated.
Course of action	"Will this step solve the stated problem?"	Practical, positive, feasible, directly addresses the problem.
Strong argument	"Is it relevant and important?"	Specific to the proposal; based on facts; not personal opinion or an emotional appeal.
Cause and effect	"Which is cause, which effect, independent or both effects of one cause?"	Judge by time order and logical dependence.

Golden rules

- Treat the statement as true and use only its information; ignore real-world knowledge unless it is common fact.
- Extreme words (always, never, only, all, none) usually make a conclusion or assumption **wrong**.
- An assumption is a hidden premise; a conclusion is a result. A conclusion can be a rewording; an assumption cannot.
- A course of action must be **realistic**; it should not promise impossible results.
- Strong argument: "Yes" arguments must give a reason; "No" arguments must reject the proposal on a real basis; both must relate to the main issue.
- Decision making: choose the answer with the best balance of short-term and long-term benefit and legality.

Statement: “Use public transport to reduce traffic jams.” Assumptions: I. Public transport is available. II. Traffic jams can be reduced. Which are implicit?

Both

Without available public transport the advice is meaningless; and the aim only makes sense if jams can be reduced.

Statement: “All students who passed the first test were given a scholarship.” Conclusions: I. Some students who got the scholarship passed. II. Only those who passed got the scholarship.

Only I

I follows if at least one student passed (and was given a scholarship). II uses “only”, which is not stated.

Problem: Many roads in the city are flooded every monsoon. Courses: I. The government should repair the drainage network. II. The people should stop living in the city.

Only I

I solves the cause. II is impractical and extreme.

Should voting be made compulsory? Arguments: I. Yes, it increases participation. II. No, because people may not like it.

Only I

II is a weak personal opinion with no reason related to the issue; I is specific and relevant.

Cause and effect

I. The price of onions rose sharply. II. Heavy rain destroyed the onion crop. Which is the cause?

II is the cause, I the effect

Crop destruction reduces supply and raises prices; the price rise cannot cause rainfall.

Logical deduction with if–then

If it rains, the match is cancelled. The match was NOT cancelled. What follows?

It did not rain

Contrapositive: not Q ⇒ not P. If rain then cancel; no cancel means no rain.

Shortcuts and tricks

- Underline the key words (only, all, must, never) before reading the options.
- For assumptions, ask “what must be true for this sentence to make sense?”
- For courses of action, prefer investigation, regulation and awareness over extreme bans or punishments.
- If the statement is in two parts, the conclusion usually links both parts.

Common mistakes

- Using your own opinion.
- Accepting a conclusion that adds information.
- Accepting strong arguments that are only emotional.

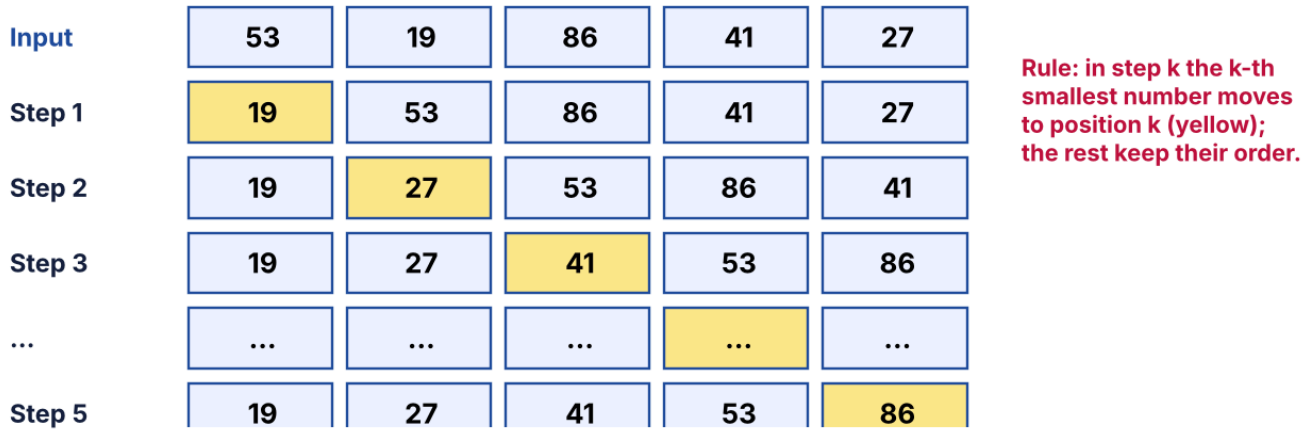
Practice set: Statement practice

1.	"If a number is divisible by 6, it is divisible by 3." 12 is divisible by 6. What follows?	Basic
2.	If A then B. B is false. What can be said about A?	Basic
3.	Statement: "Do not leave the house; the weather is stormy." Is "the speaker cares about safety" an assumption?	Medium
4.	If all engineers are graduates and some graduates are managers, can we conclude all engineers are managers?	Medium
5.	Cause–effect: I. Fuel prices rose. II. Bus fares were increased. Which is the effect?	Medium
6.	Statement: "Only exercise improves health." Is this a safe conclusion from the statement "Exercise improves health"?	Advanced
7.	If P then Q; if Q then R. P is true. What follows?	Advanced

14 Input–Output and Machine Logic

REASONING ABILITY · PART B · TCS FAVOURITE

A machine rearranges a list of words or numbers step by step. You study the first one or two steps to discover the **rule**, then predict a later step or the final arrangement. The rule is usually one of: **sorting** (ascending or descending, one element per step), **alternate positions**, **reversing groups**, or **arithmetic on selected items**.



A typical machine. Input: 53 19 86 41 27. In each step the next smallest number is brought to the next position. (Output generated by code.)

Rule-finding checklist

- Compare input and step 1: which element moved, and where? That tells you the rule.
- Compare step 1 and step 2: the **same rule repeats** on the remaining part.
- Sorting types: ascending/descending by value, by alphabet, by number of letters, by last digit, by even-before-odd.
- Alternate moves: smallest to the left then largest to the right; or odd positions then even positions.
- The number of steps needed to finish a sort of n items is at most n − 1 for these machines.
- Questions often ask: “Which step is the last?” or “What is the position of X in step k?” Compute the whole table, then read.

Input: 53 19 86 41 27. Step 1: 19 53 86 41 27. Step 2: 19 27 53 86 41. Find step 3.

19 27 41 53 86

Rule: in step k the k-th smallest number goes to position k; the rest keep their order. Step 3 brings 41 to position 3: 19 27 41 53 86.

Input: sun moon star sky cloud. Step 1: cloud sun moon star sky. Find the last step.

cloud moon sky star sun

Alphabetical order, one word per step: step 1 brings cloud first, step 2 brings moon, step 3 brings sky, step 4 brings star. Final: cloud moon sky star sun. (Verified by code.)

Shortcuts and tricks

- Rewrite the first two steps on rough paper side by side; the rule appears.
- Work with positions not values to see which block moved.
- When the rule involves arithmetic, check it for two numbers before applying.
- A reversal of the entire list is rare; reversal of a block is common.

Common mistakes

- Guessing the rule from one step only.
- Mixing ascending and descending in alphabetical orders.
- Skipping a step to save time and losing the intermediate order.

Practice set: Machine practice

1.	Step 1 of a machine on 8 3 9 5 gives 3 8 9 5. What does step 2 give if the rule is “bring the next smallest to the next position”?	Basic
2.	Input: 42 15 63 27. If the rule moves the largest to the front each time, step 1?	Basic
3.	Sort “dog cat bat ant” alphabetically one word per step. Step 2 gives?	Medium
4.	Input 5 4 3 2 1. If each step swaps the first two, step 1?	Medium
5.	Machine reverses the order of the first k elements in step k. Input 1 2 3 4, step 3?	Advanced
6.	Machine adds 2 to each number in odd positions. Input 4 7 9 2, step 1?	Advanced
7.	Machine arranges numbers by increasing sum of digits. Input 65 12 33 40. Result?	Advanced

REASONING ABILITY · PART B · ADVANCED

15 Logical Puzzles, Counting and Cryptarithms

This chapter collects the “brain-teaser” formats: truth-teller / liar puzzles, jug measures, cryptarithms (letters standing for digits) and counting figures. They reward **systematic case work** over cleverness.

1. Truth-tellers and liars

A says “B is a liar”. B says “A and I are of the same type”. Each person is either always truthful (T) or always a liar (L). What are A and B?

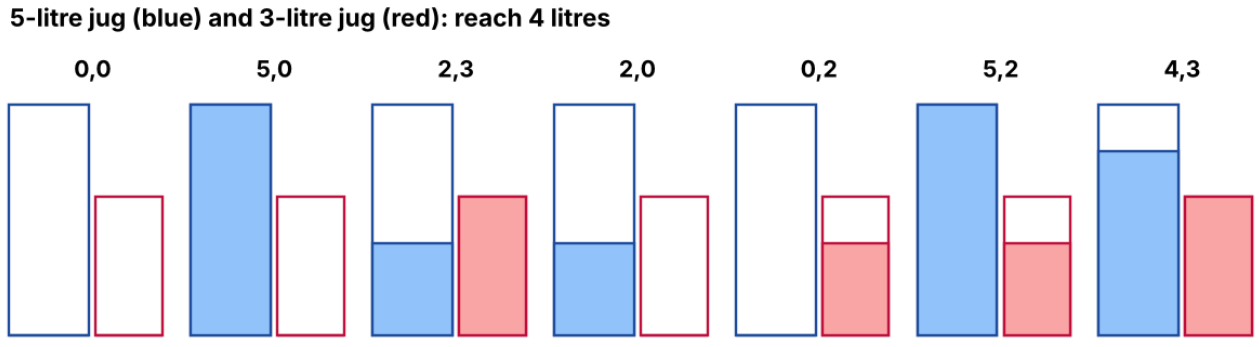
A truthful, B liar

Try all four cases: only A = T, B = L is consistent. If A = T then B is a liar; B’s claim “same type” is false, which is correct for a liar. (Checked by exhaustive enumeration.)

Truth-teller method

- List the unknowns (T or L for each person); with n persons there are 2^n cases.
- Test each case: a T’s statement must be true; an L’s statement must be false.
- Exactly one case usually survives. Use symmetry to cut work (if two persons say the same thing they are of the same type).

2. Jug problems



State = (litres in 5-litre jug, litres in 3-litre jug). Shortest sequence found by search.

With a 5-litre and a 3-litre jug and unlimited water, how can you measure exactly 4 litres?

See sequence

Fill 5 → pour into 3 (5-jug has 2) → empty 3 → pour 2 into 3 → fill 5 → pour into 3 until full (1 litre needed) → 5-jug now has 4 litres. Verified by shortest-path search.

3. Cryptarithm

SEND + MORE = MONEY, each letter a different digit, S and M non-zero. Find S + E + N + D.

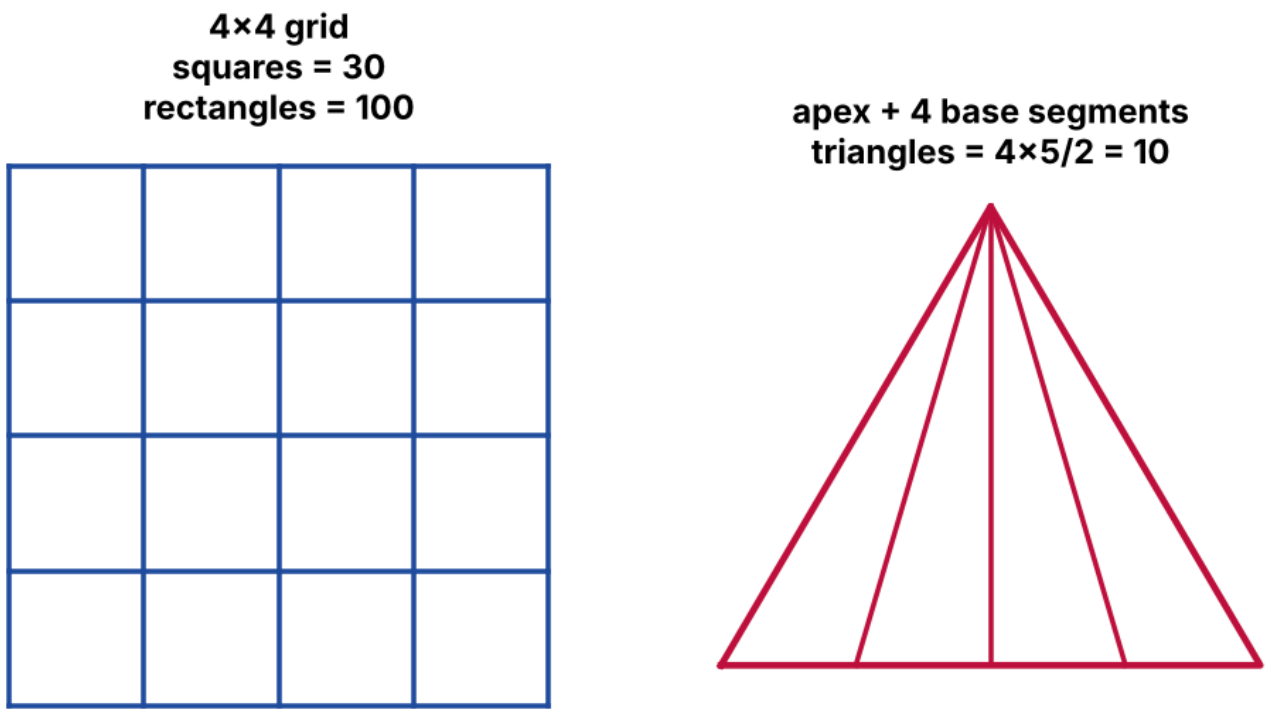
27

Solution (unique, found by checking all arrangements): S = 9, E = 5, N = 6, D = 7, M = 1, O = 0, R = 8, Y = 2. 9567 + 1085 = 10652. Sum = 9 + 5 + 6 + 7 = 27.

Shortcuts and tricks

- In a cryptarithm start with the leftmost carry: M is the carry of two 4-digit sums, so M = 1.
- Use the units-digit column and parity to restrict letters.
- Write the candidate digits next to each letter and cross them out.

4. Counting figures



Squares and rectangles in a grid and triangles through an apex.

Counting formulas

- Rectangles in an m × n grid: **[m(m+1)/2] × [n(n+1)/2]** (4×4 gives 100).
- Squares in an m × n grid: **Σ (m − k + 1)(n − k + 1)** for k = 1...min(m,n) (4×4 gives 30).
- Triangles with a common apex and n points on the base: **n(n + 1)/2**.
- Number of straight lines through n points (no 3 collinear): **n(n − 1)/2**.
- Number of diagonals of an n-gon: **n(n − 3)/2**.
- Diagonals of a hexagon = 9; an octagon = 20.

How many squares and rectangles (including squares) are in a 3 × 4 grid?

20 squares; 60 rectangles

Squares: 12 + 6 + 2 = 20. Rectangles: (3×4/2)(4×5/2) = 6 × 10 = 60. Verified by code.

Shortcuts and tricks

- Count in layers (size 1, size 2, ...) to avoid repeats.
- For triangles inside a figure, label the points and list triplets of lines.
- Use symmetry to count part of the figure and multiply.

Common mistakes

- Counting the same square or triangle twice.
- Forgetting that all squares are rectangles when the question says “rectangles”.

Practice set: Puzzle and counting practice

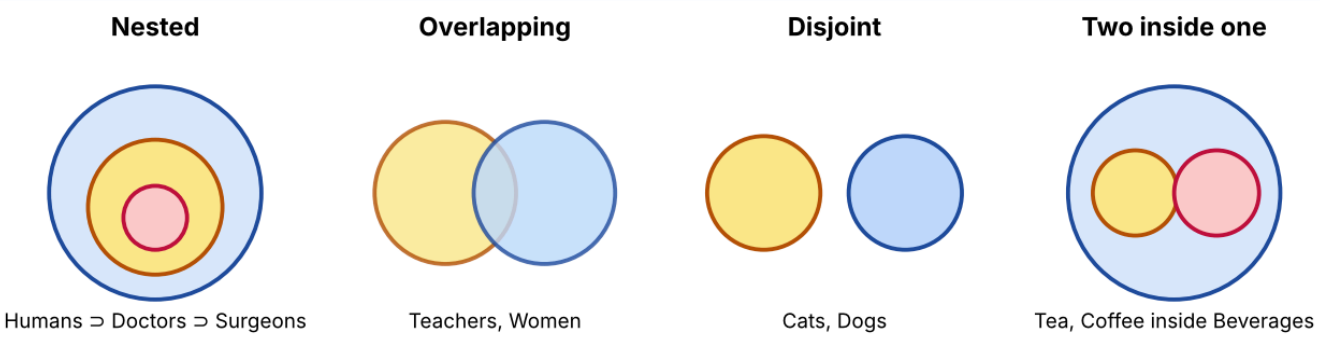
1.	Number of diagonals in a decagon (10-gon)?	Basic
2.	How many rectangles are in a 2 × 3 grid?	Basic
3.	A says “I am a liar”. Can A be truthful?	Medium
4.	Triangles with a common apex and 6 points on the base?	Medium
5.	How many squares in a 5 × 5 grid?	Medium
6.	If 4 persons shake hands in a group, each exactly once, how many handshakes?	Advanced
7.	Number of straight lines from 8 points with no three collinear?	Advanced

REASONING ABILITY · PART B · SCORING

16 Venn Diagrams and Set Problems

Two types of questions appear: **(1) choose the Venn diagram that represents a group of items** (relationship questions) and **(2) count people in regions** of a two- or three-set diagram. The first is pure logic; the second is inclusion–exclusion.

1. Relationship diagrams

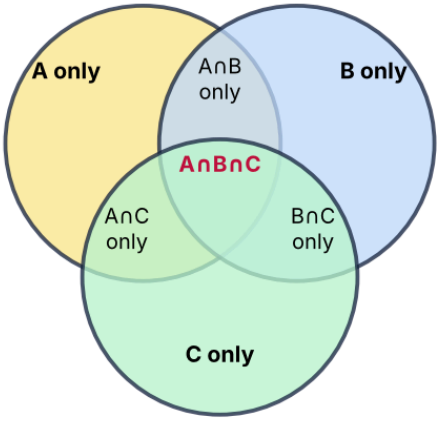


Four basic patterns. Nested: each is a part of the next. Overlapping: some common members. Disjoint: no common member.

Items	Diagram	Why
Humans, Doctors, Surgeons	Nested	All surgeons are doctors; all doctors are humans.
Teachers, Women, Mothers	Teachers and women overlap; mothers inside women	A teacher may or may not be a woman; all mothers are women.
Cat, Dog, Animals	Cat and dog disjoint, both inside animals	No cat is a dog; both are animals.
Pen, Pencil, Stationery	Pen and pencil disjoint inside stationery	Two different things, same category.
Rivers, Water, Ganga	Ganga inside rivers; rivers inside or overlapping water	Ganga is a river; rivers hold water.

2. Counting regions

Three-circle Venn diagram: 7 regions + outside




```
|A∪B∪C| = |A|+|B|+|C|
       − |A∩B| − |B∩C| − |A∩C|
       + |A∩B∩C|

Example: |A|=30, |B|=25, |C|=20
|A∩B|=10, |B∩C|=8, |A∩C|=7, all three = 4
Union = 75 − 25 + 4 = 54

Only A = |A| − |A∩B| − |A∩C| + |A∩B∩C|
        = 30 − 10 − 7 + 4 = 17
Exactly two sets = Σ(pairs) − 3×(all three)
        = 25 − 12 = 13
```

Inclusion–exclusion for three sets. The worked example is on the right.

Formulas

- Two sets: $|A \cup B| = |A| + |B| - |A \cap B|$.
- Three sets: $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$.
- Only A = $|A| - (A \cap B) - (A \cap C) + (A \cap B \cap C)$.
- Exactly two sets = Σ pairwise − 3 × (all three).
- Neither = Total − Union.
- Number who like at least one of the sets = Union.

In a class, 30 like cricket, 25 like football, 20 like tennis, 10 like both cricket and football, 8 football and tennis, 7 cricket and tennis, 4 like all three. How many like at least one game?

54

30 + 25 + 20 − 10 − 8 − 7 + 4 = 54. (Computer checked.)

From the data above, how many like cricket only?

17

30 − 10 − 7 + 4 = 17.

In a group of 100, 60 like tea and 50 like coffee, 20 like neither. How many like both?

30

Union = 100 − 20 = 80. Both = 60 + 50 − 80 = 30.

Shortcuts and tricks

- Always fill the centre (all three) first, then the pairs, then the singles.
- Subtract the triple overlap from each pair before computing “only” regions.
- Check by adding all region values; it must equal the total.
- In relationship questions, check “all” (nested) before “some” (overlapping) before “none” (disjoint).

Common mistakes

- Reading “A and B” as “A only and B only” instead of the common part.
- Forgetting the outside region (“neither”).

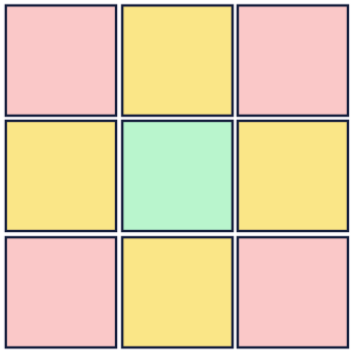
Practice set: Venn practice

1. Which diagram: Poets, Writers, Humans? — Basic
2. In a group of 70, 40 read The Hindu, 35 read the Times, 10 read neither. How many read both? — Basic
3. If 45 like A, 38 like B and 12 like both, how many like at least one? — Medium
4. With the data A=30, B=25, C=20, AB=10, BC=8, AC=7, ABC=4, how many like exactly two? — Medium
5. Same data: how many like B only? — Medium
6. Same data: how many like C only? — Advanced
7. From a total of 100 persons with the same data, how many like none? — Advanced

Cubes and Dice

Cube questions come in two forms: **painted cube cut into small cubes** (counting by number of coloured faces) and **dice** (which face is opposite which). Both are solved with fixed formulas and a small amount of imagination.

Top layer of a 3×3×3 cube
corner(3 faces), edge(2), face(1)



Cube of side n, cut into n^3 small cubes

3 faces coloured: 8 (corners)
2 faces coloured: 12 (n − 2)
1 face coloured: 6 $(n-2)^2$
0 faces coloured: $(n-2)^3$

n = 4: 8, 24, 24, 8
n = 5: 8, 36, 54, 27

Other cube facts

Edges: 12 Faces: 6 Vertices: 8
Face diagonal = $a\sqrt{2}$
Space diagonal = $a\sqrt{3}$
Surface area = $6a^2$
Volume = a^3
Cut into k × k × k pieces: k^3 pieces
Cuts needed along one direction: k − 1

Left: the top layer of a 3×3×3 cube. Corners have 3 painted faces, edges 2, face centres 1, and the inside cube 0.

Painted cube formulas (n small cubes along one edge)

- 3 faces painted: always **8** (corners).
- 2 faces painted: **12(n − 2)** (edge pieces).
- 1 face painted: **$6(n-2)^2$** (face pieces).
- No face painted: **$(n-2)^3$** (inside).
- Total = n^3. Check: $8 + 12(n-2) + 6(n-2)^2 + (n-2)^3 = n^3$.
- If a cube is painted on only some faces, count the pieces on those faces and subtract overlaps.
- Cut-planes: a cube cut by a, b, c planes parallel to three directions gives (a+1)(b+1)(c+1) pieces.

A cube of side 5 cm is painted on all faces and cut into 1 cm cubes. How many have exactly 2 faces painted?

36

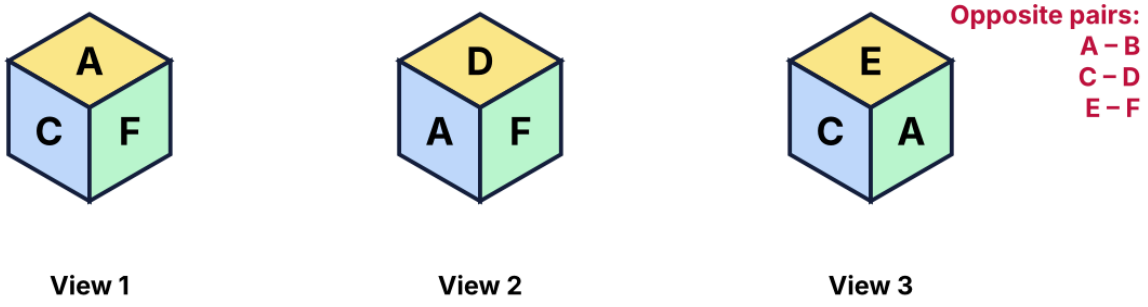
12(n − 2) = 12 × 3 = 36. Computer enumeration of all 125 cubes agrees (8, 36, 54, 27).

A 4 × 4 × 4 cube is painted on all six faces and cut. How many small cubes have no paint?

8

$(4-2)^3 = 8$.

Dice



Three views of the same die. Letters are on the faces. The top face is shown as the top rhombus, the left rhombus is the front and the right rhombus is the right face.

Dice rules

- Standard die: opposite faces add to **7** (1–6, 2–5, 3–4).
- Two faces in the same view are **adjacent**, never opposite.
- If two views share a common face, the faces that are NOT shared appear around it in rotation order; keep the rotation direction the same.
- Three views are usually enough to find all opposite pairs. A face never appears in one view with its opposite.
- A face that appears in several views is adjacent to every face it appears with. A cube has only four neighbours per face, so the one remaining face is the opposite.

Three views of the same die are shown above (A, C, F), (D, A, F), (E, C, A). Which letter is opposite A?

B

A appears in all three views, next to C, F (view 1), D, F (view 2) and E, C (view 3). So A is adjacent to C, D, E and F, and the only face left, B, is opposite A. Checked by trying all 720 labellings: the only consistent opposite pairs are A–B, C–D, E–F. So B is opposite A.

Shortcuts and tricks

- List which faces ever appear together; faces that never appear together are opposite candidates.
- The face common to two views stays in the same place; the others rotate about it.
- For dice with numbers, use the 7-sum rule immediately.
- If a face never appears with another face in any view, they are candidates for being opposite.

Common mistakes

- Treating two faces seen together as opposite.
- Ignoring the direction of rotation in two views.

Practice set: Cube and dice practice

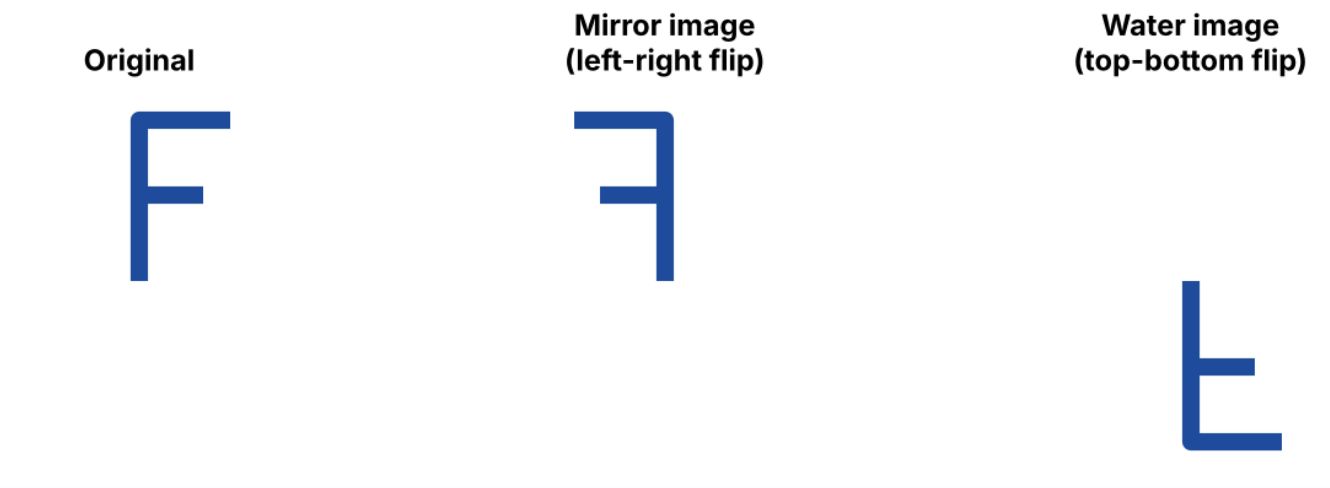
1. A cube of side 6 is painted on all faces and cut into unit cubes. How many have exactly one face painted? **Basic**

2.	Same cube: how many have three faces painted?	Basic
3.	Same cube: how many have no paint?	Medium
4.	A standard die shows 2 on top. What is on the bottom?	Medium
5.	A 5×5×5 cube painted outside is cut into unit cubes. Number with 2 faces painted?	Medium
6.	A cube painted on two opposite faces only, side 4, is cut into unit cubes. How many have two faces painted?	Advanced
7.	A 3×3×3 cube is cut by one plane in each direction (3 planes). How many pieces?	Advanced

REASONING ABILITY · PART B · VISUAL

18 Mirror, Water Image, Paper Folding and Cube Nets

Four visual topics that share one idea: **a flip, a fold or a rotation keeps distances but changes left/right or up/down**. Learn the rule for each and apply it to one letter at a time.



Letter F: the mirror flips left to right (the vertical line stays on the other side) and the water image flips top to bottom.

Mirror and water image

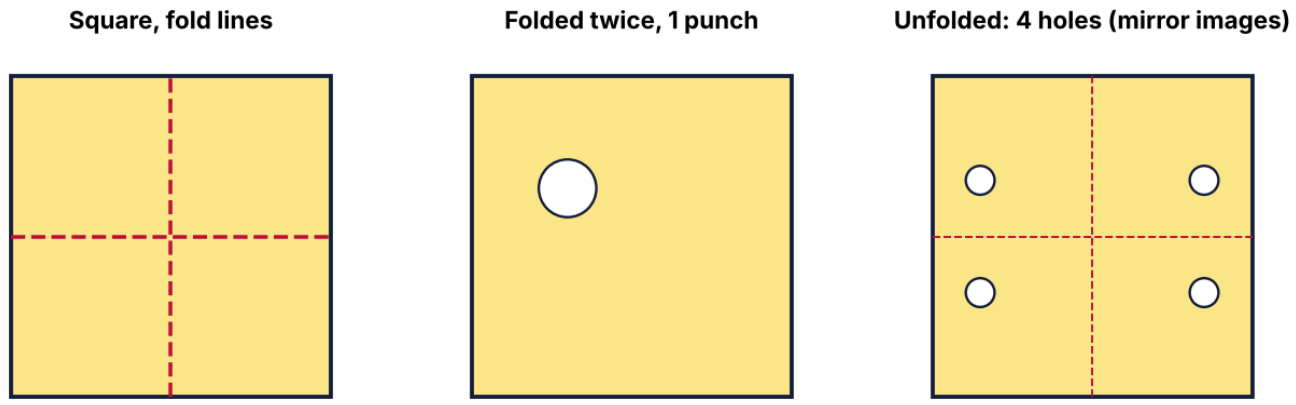
- **Mirror image** (mirror placed to the right or left): left and right are swapped; top and bottom stay. The order of letters/digits in a word reverses.
- **Water image** (mirror placed below, as in water): top and bottom are swapped; left and right stay. Word order stays.
- Letters with the same mirror image: **A H I M O T U V W X Y**. Letters with the same water image: **B C D E H I K O X**.
- Digits with their own mirror: 0, 1, 8. Digits with their own water image: 0, 3, 8.
- Clock mirror image: 11:60 − time (see Clock chapter).
- Mirror image of a calendar date/number: write it backwards and reverse each character.

Which of these words looks identical in its mirror image: MAN, TOT, TAB, BAT?

TOT

Reverse the order (TOT stays TOT) and flip every letter: T, O, T are all left-right symmetric. In MAN the letter N is not symmetric, and B is not symmetric.

Paper folding and cutting



A hole punched on a paper folded twice becomes 4 holes after unfolding, placed symmetrically about both fold lines.

Folding rules

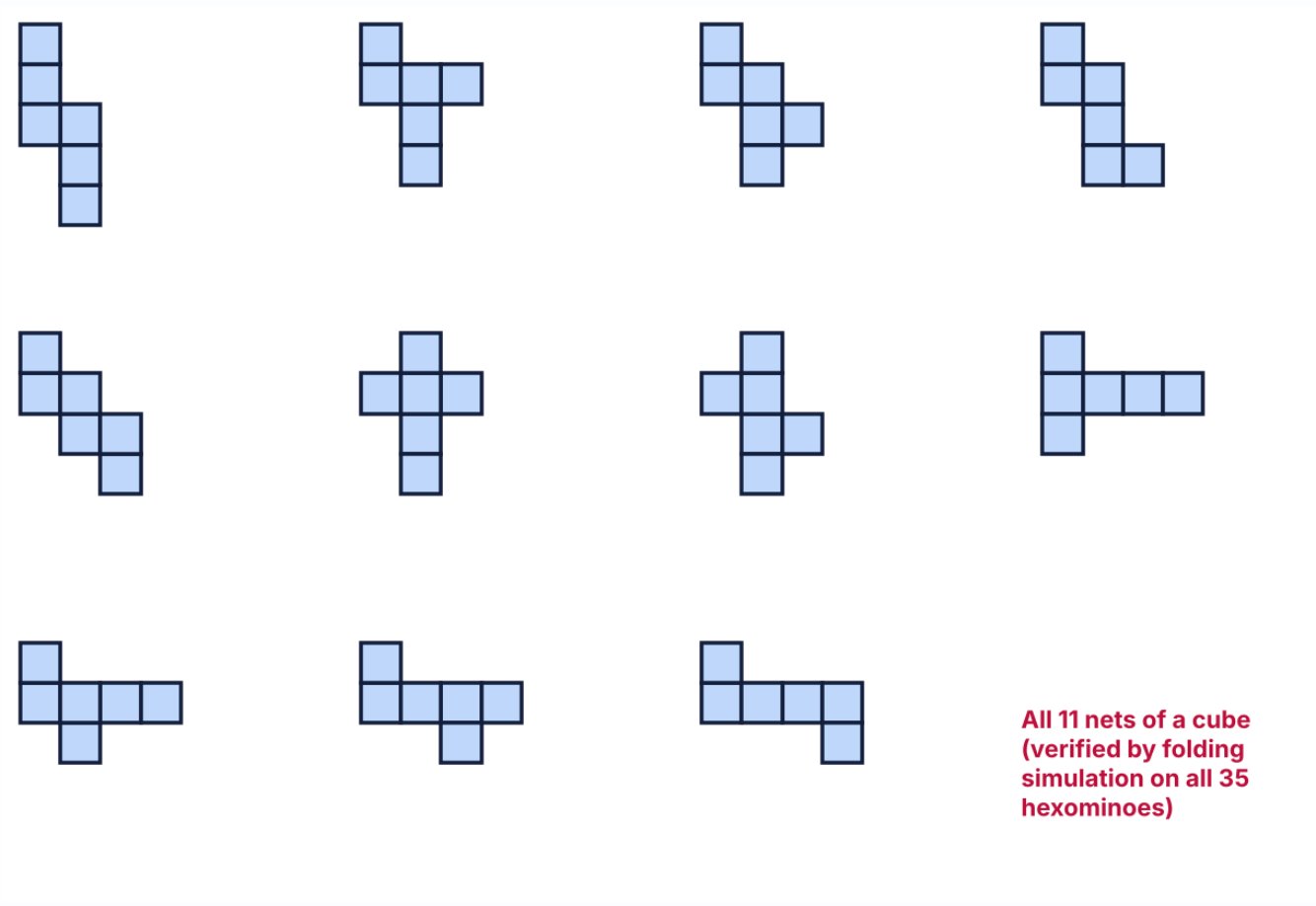
- Each fold doubles the number of layers; a hole through a paper folded k times (away from the fold line) gives **2^k holes**.
- A hole punched on a fold line gives half as many holes (the hole is cut in two halves).
- Unfold in reverse order of folding; each unfolding mirrors every mark across that fold line.
- For a diagonal fold, the mirror line is the diagonal; marks reflect across it.

A square paper is folded in half twice and a circular hole is punched near its centre of the folded square (away from the fold lines). How many holes appear on unfolding?

4

Two folds give four layers: $2^2 = 4$ holes. They are mirror images in both fold lines.

Cube nets



The 11 nets that fold into a cube (generated and checked by a folding simulation).

Net rules

- Exactly **11** flat nets fold into a cube.
- A net with **four squares in a line** and one on each side of the line always works (6 of the 11 nets).
- Two opposite faces are separated by exactly one square in any straight strip of three squares.
- If any net contains a 2 × 2 block of squares, it will NOT fold into a cube.
- A strip of more than four squares in a row cannot fold into a cube.

Does a net with a 2 × 2 block of squares fold into a cube?

No

The four squares of the block cannot all meet at one corner of the cube without overlap. (All 35 hexominoes were tested by simulation: those with a 2 × 2 block fail.)

Shortcuts and tricks

- Do one letter at a time for mirror/water images.
- For paper folding, mark the hole position, then mirror it across each fold line in reverse order.
- For nets, find opposite faces first (one square apart in a strip), then match shapes.

Common mistakes

- Using the mirror image where the question asks for the water image.
- Forgetting to reverse the order of the letters in a word for a mirror image.

Practice set: Visual practice

1. Which letters look the same in a mirror: M, B, T, K? — Basic
2. A square paper is folded 3 times and a hole is punched away from the fold lines. How many holes? — Basic
3. A hole is punched on a fold line of a paper folded once. How many holes on unfolding? — Medium
4. Does a straight strip of 5 squares plus one square fold into a cube? — Medium
5. How many nets of a cube are there? — Medium
6. Water image of the word BOX? — Advanced
7. How many layers of paper after 4 folds? — Advanced

REASONING ABILITY · PART B · VISUAL

19 Non-Verbal Reasoning: Figure Series, Matrix and Pattern

Figure questions have no words. They test whether you can spot **what changes** from one picture to the next: position, rotation, size, number of parts, shading. Always check changes in **this order**: (1) number of elements, (2) position, (3) rotation or reflection, (4) shading, (5) size.



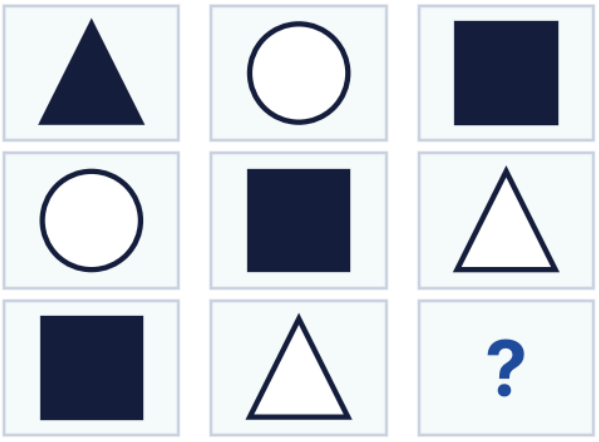
The dot moves one corner clockwise each step: TL → TR → BR → BL → TL → ? (answer: top-right)

Movement of one element: one corner clockwise at each step. Next figure: dot at the top-right.

Change checklist

- **Rotation:** an arrow may turn 45°, 90° or 180° clockwise or anticlockwise each step; measure two steps to fix the angle.
- **Movement:** an element moves a fixed number of cells or corners each time, often clockwise.
- **Count change:** the number of lines, dots or sides increases by 1 or by 2 (or alternates).
- **Shading:** alternates every step, or shades one more part each step.
- **Odd–even rule:** if the odd-numbered and even-numbered figures follow separate patterns, solve the two series independently.
- **Mirror/flip:** figures alternate between an original and its mirror image.

Matrix completion



Each row and each column contains all three shapes once (Latin square). The shading alternates with (row + column).

Missing cell:
shape index (2+2)%3 = 1 → circle
(2+2)%2 = 0 → filled circle

A 3 × 3 matrix: each shape appears once in each row and column.

A 3 × 3 figure matrix has triangle/circle/square in each row and column (Latin square) and the shading alternates with (row + column). Which figure goes in the bottom-right cell?

Filled circle

Row 3, column 3: shape index (2 + 2) mod 3 = 1, the circle; shading uses (2 + 2) mod 2 = 0, filled.

Other formats

Format	How to attack
Analogy figures (A : B :: C : ?)	Find the transformation A to B (rotate, add, remove, flip), apply to C.
Odd figure out	Count sides/parts/shaded areas in each; one breaks the common property.
Embedded figures	Check the hidden figure in each option by tracing lines; use the distinctive angle first.
Pattern completion	Match the missing part against the break in lines (and edge continuity).
Counting triangles/squares	Use the counting formulas of the Logical Puzzles chapter.
Dot situation	Place dots in the regions that are common to the required shapes (circle and square, etc.).

An arrow points North, then East, then South in consecutive figures. What next?

West

90° clockwise each step: N → E → S → W.

Squares with 1, 3, 5 dots in consecutive figures. How many dots in the 5th?

9

Arithmetic progression with difference 2: 1, 3, 5, 7, 9.

Shortcuts and tricks

- Compare figure 1 with figure 3 as well as figure 1 with figure 2; alternating patterns show up.
- Mark each element by letter on the page and track it.
- Eliminate options with the wrong count first.
- For rotating figures remember that the rotation angle can accumulate (45°, 90°, 135°...).

Common mistakes

- Looking at shapes only and ignoring shading or orientation.
- Assuming the pattern has a single rule when it has two (position and shading).

Practice set: Figure practice

1. An arrow points West, then North, then East. What next? — Basic
2. A square rotates 45° clockwise each step. After 4 steps, what is the total rotation? — Basic
3. Triangles have 1, 2, 4, 8 dots. How many in the next? — Medium
4. A figure has 3, 6, 10, 15 lines. How many in the next? — Medium
5. A black dot moves two corners clockwise each step starting at top-left. Where after 3 steps? — Medium
6. A shape alternates between shaded and unshaded and also rotates 90° each step. In the 6th figure, which state? — Advanced

7. A 3 × 3 matrix has the row sums of dots equal to 12. Row 1 has 3, 4, 5. Row 2 has 6, 2, 4. What is the sum of row 3? **Advanced**

REASONING ABILITY · FINAL CHAPTER

20 Master Strategy, Formula Sheet and Mini Test

This last chapter gives you a **study plan**, a one-page **formula sheet** and a **20-question mini test** that mixes every topic. Take the test timed (30 minutes) and compare with the answer key.

How to study this guide

Week	Topics	Daily target
1	Series, Coding, Blood relations, Directions	2 topics a day, 15 practice questions each
2	Seating, Puzzles, Syllogism, Clock/Calendar	Full chapter + 20 questions
3	Ranking, Inequality, Analogy, Data sufficiency, Statements	Speed drills, 10 minutes each
4	Machine, Logic, Venn, Cubes, Visual	Timed sets of 10 questions
5	Full mocks; review mistakes	1 mock per day, then an error log

Time-management rules for the exam

- Spend at most **90 seconds** on any reasoning question the first time; mark and return.
- Do the quick topics first: series, coding, ranking, direction, clock, calendar, inequality, analogy.
- Attempt seating and puzzles after the quick set; draw large diagrams.
- Never leave a question half-drawn: either finish it or move on.
- If options are given, eliminate by one fixed condition first.
- Keep the last 5 minutes for revisiting marked questions.

One-page formula master sheet

Topic	Key facts
Series	AP: $a + (n-1)d$; GP: $a\,r^{n-1}$; squares, cubes, primes, Fibonacci; alternate series
Coding	Letter ±shift; opposite letter = 27 − p; sum of positions; EJOTY
Blood relation	Mother's brother = maternal uncle; father's sister = paternal aunt; draw the tree
Direction	$d = \sqrt{(x^2+y^2)}$; triples 3-4-5, 5-12-13, 8-15-17; right turn = clockwise
Seating	Total = left + right − 1; between = b − a − 1; circle n people: $(n-1)!$ ways
Puzzles	Knockout n − 1; round robin $n(n-1)/2$; handshakes $n(n-1)/2$
Syllogism	All+All = All; Some+All = Some; All+No = No; Some+Some = none
Clock	$\lvert 30H - 5.5M \rvert$; overlap 60H/11; mirror 11:60 − T; 22 overlaps/24 h
Calendar	Ordinary 1 odd day, leap 2; 100 y = 5, 200 y = 3, 300 y = 1, 400 y = 0
Ranking	Total = top + bottom − 1; between = difference − 1
Inequality	Combine like signs; > with < gives nothing
Venn	$\lvert A \cup B \cup C \rvert = \Sigma A - \Sigma \text{pairs} + \text{all three}$
Cubes	8 corners; $12(n-2)$; $6(n-2)^2$; $(n-2)^3$; die opposite sum 7
Counting	Diagonals $n(n-3)/2$; lines $n(n-1)/2$; rectangles $C(m+1,2)\cdot C(n+1,2)$
Paper	k folds ⇒ 2^k holes; mirror flips L–R, water flips T–B

Mini test (20 questions, 30 minutes)

Mixed mini test

1. Series: 7, 14, 28, 56, ?
2. Coding: if A = 1, B = 2, ..., find the sum of values of the letters of DOG.
3. A woman points to a man and says “He is the son of my mother’s only son.” How is the man related to her?
4. A man walks 6 m North and then 8 m East. Distance from the start?
5. In a row of 30, X is 12th from the left. Position from the right?
6. All A are B, All B are C. Does “All A are C” follow? (Yes/No)
7. Angle between the hands at 5:30 (degrees)?
8. Odd days in 200 years?
9. Ravi is 17th from the top and 25th from the bottom. Class size?
10. P > Q ≥ R = S. Does P > S follow? (Yes/No)
11. 6 : 216 :: 9 : ?
12. In a group of 100, 60 like tea, 50 like coffee, 20 like neither. How many like both?
13. A cube of side 6 is painted on all faces and cut into unit cubes. How many have exactly one painted face?
14. Number of rectangles in a 3 × 3 grid?
15. Knockout tournament with 16 teams: total matches?
16. Round robin with 7 teams: total matches?
17. Diagonals of a 12-sided polygon?
18. Minutes past 4 at which the hands first overlap (as a mixed fraction)?
19. A square paper is folded twice and one hole is punched away from the fold lines. Holes after unfolding?
20. Mirror image of the time 3:25 on a clock?

Answer key

Answers to every practice set and to the mini test. B = Basic, M = Medium, A = Advanced.

1. Series

1. 158 (B)
2. 169 (M)
3. L (gaps 2,3,4,5) (M)
4. 210 (n^3-n) (A)
5. N47T (A)
6. 121 (M)
7. EV (B)
8. 63 (B)

2. Coding-Decoding

1. JQWUG (+2) (B)
2. 26 (B)
3. NPPO (+1) (B)
4. GIZRM (M)
5. DQOD (B)
6. ro (M)
7. NKQH (M)
8. 1 (A)

3. Blood Relations

1. Paternal grandmother (B)
2. Maternal cousin (cousin) (M)
3. Paternal uncle (M)
4. Maternal uncle (M)
5. Sister-in-law (M)
6. Son (A)
7. Mother-in-law (A)

4. Directions

1. 13 m (B)
2. East (B)
3. 4 km (direction North) (M)
4. 15 m (M)
5. North (facing West, right = North) (M)
6. 41 m; North-West (A)
7. 3 m East (A)

5. Seating Arrangements

1. 15th (B)
2. 7 (B)
3. 3 (M)
4. 11 (M)
5. 120 (M)
6. 0 (they are adjacent) (A)
7. 2 (P Q S T R and P Q T S R) (A)

6. Puzzles

1. 63 (B)
2. 36 (B)
3. D smallest, C largest (M)
4. 56 (M)
5. 0 (they are adjacent) (M)
6. D oldest, C youngest (A)
7. 42 (A)

7. Syllogism

1. No (B)
2. Yes (B)
3. No (possible only) (M)
4. Yes (M)
5. Yes (M)
6. Yes (A)
7. Neither (both are only possible) (A)

8. Clock and Calendar

1. 15° (B)
2. 172.5° (B)
3. 22 (M)
4. 3:40 (M)
5. Friday (M)
6. 0 (M)
7. Monday (A)

9. Ranking and Alphabet

1. 29 (B)
2. O (B)
3. S (M)
4. 12 (M)
5. 0 (M)
6. 3 pairs: (A,H), (G,M), (R,T) (A)
7. 72 (18+5+1+19+15+14) (A)

10. Inequalities

1. Yes (B)
2. No (B)
3. 16 ($18 - 4 \times 3 \div 6$) (M)
4. Both follow (M)
5. 12 (M)
6. P < R yes; Q > S no (A)
7. I and II only (A)

11. Analogy

1. Cut (B)
2. Gallery (B)
3. 64 (M)
4. DW (M)
5. 9 (M)
6. HJL (A)
7. 220 (A)

12. Data Sufficiency

1. Both together (C) (B)
2. Both (C) (B)
3. Not sufficient (E) (M)
4. Either alone (D) (M)
5. Both together (C) (M)
6. Not sufficient: 13 and 31 (E) (A)
7. Both (C) (A)

13. Statement-based

1. 12 is divisible by 3 (B)
2. A is false (B)
3. Yes (M)
4. No (M)
5. II (M)
6. No (“only” is added) (A)
7. R is true (A)

14. Input–Output

1. 3 5 8 9 (B)
2. 63 42 15 27 (B)
3. ant bat dog cat (M)
4. 4 5 3 2 1 (M)
5. 3 1 2 4 (step 1: 1 2 3 4, step 2: 2 1 3 4) (A)
6. 6 7 11 2 (A)
7. 12 40 33 65 (A)

15. Logical puzzles

1. 35 (B)
2. 18 (B)
3. No (paradox) (M)
4. 21 (M)
5. 55 (M)
6. 6 (A)
7. 28 (A)

16. Venn diagrams

1. Nested: Poets ⊂ Writers ⊂ Humans (B)
2. 15 (B)
3. 71 (M)
4. 13 (M)
5. 11 (M)
6. 9 (A)
7. 46 (A)

17. Cubes and Dice

1. 96 (B)
2. 8 (B)
3. 64 (M)
4. 5 (M)
5. 36 (M)
6. 0 (A)
7. 8 (A)

18. Image, Paper, Nets

1. M and T (B)
2. 8 (B)
3. 1 (a single hole, cut in half on each side) (M)
4. No (M)
5. 11 (M)
6. BOX (B, O, X are all symmetric top-to-bottom) (A)
7. 16 (A)

19. Non-verbal

1. South (B)
2. 180° (B)
3. 16 (M)
4. 21 (M)
5. Bottom-right (TL → BR → TL → BR) (M)
6. Unshaded and turned 90° clockwise from the start (5 turns = 450°) (A)
7. 12 (A)

Mini test (20 questions)

1. 112 (M)
2. 26 (M)
3. Nephew (her brother’s son) (M)
4. 10 (M)
5. 19 (M)
6. Yes (M)
7. 15 (M)
8. 3 (M)
9. 41 (M)
10. Yes (M)
11. 729 (M)
12. 30 (M)
13. 96 (M)
14. 36 (M)
15. 15 (M)
16. 21 (M)
17. 54 (M)
18. 21 9/11 (M)
19. 4 (M)
20. 8:35 (M)